

1 Light-sharing control for tracker-based agrivoltaics
2 under a yield floor: a simulation study of what
3 learning adds and what shade-response uncertainty
4 costs

5 Quang-Vinh Dang*

6 *British University Vietnam, Hung Yen, Vietnam*

7 **Abstract**

Single-axis trackers can shift their daily rotation from maximum electricity towards letting light through (light sharing). We quantify its value under a 90% yield-retention floor with a differentiable simulator. On 92 held-out crop–site pairs and the proportional-growth shade response, light-sharing rules allowed arrays 2.1–2.9 times as dense as static arrays at equal mean retention and raised the land-equivalent ratio (LER) from 1.07 to 1.18–1.24, with the floor met in 69–76% of seasons (static arrays: 93%). The crop response to shade is the least certain input: a hierarchical model fitted to 428 observations from 58 studies predicts a new study of the four staple crop types (138 observations, 30 studies) only to about 0.22 in relative yield, no better than proportional loss, so we built ensembles of plausible responses from the learned curves and field trials. Within these ensembles, a density chosen under the mean response left about 70% of pairs at the floor when another plausible response held, against 85–87% for an oracle design with the true response; a floor raised by 0.02 and an ensemble-quantile rule were not distinguishable at equal compliance, and neither was safe outside the ensemble, whose spread is narrower than the between-study spread. Pooled over three splits (76 pairs), a neural controller initialised from the written phenology rule beat it by 0.044 LER but a CMA-ES-optimised window, re-tuned on each split, by only 0.007 (95% interval 0.002–0.013; 0.004 at matched retention, interval including zero); ensemble or biomass-signal training did not help within our budget. Under illustrative costs, light sharing breaks even with the open field at floor-designed densities at 70–92 USD MWh^{−1}

*Corresponding author.
Preprint submitted to Artificial Intelligence in Agriculture
Email address: vinh.dq4@buv.edu.vn (Quang-Vinh Dang)

(static arrays at 59) and a conventional plant is more profitable above 48. For maize and soybean (53% of the upscaled 14 PWh yr⁻¹) the simulator's bias is uncertain.

8 *Keywords:* Agrivoltaics, Light sharing, Solar tracking, Food–energy nexus,
9 Crop modelling, Differentiable simulation, Decision under uncertainty

10 1. Introduction

11 Two of the defining challenges of this century compete for the same re-
12 source: land. The energy transition requires terawatts of new solar pho-
13 tovoltaic (PV) capacity, and ground-mounted solar farms need roughly one
14 to two hectares per megawatt. Food production must rise while crops are
15 exposed to more frequent heat waves and droughts. Warming alone is pro-
16 jected to reduce global yields of maize, wheat, rice and soybean by several
17 per cent per degree (Zhao et al., 2017; IPCC, 2021), and a new generation
18 of climate and crop models shows these impacts emerging earlier than pre-
19 viously expected (Jägermeyr et al., 2021). Croplands are also where solar
20 power potential is greatest (Adeh et al., 2019). They are flat, sunny, acces-
21 sible and close to rural electricity demand, so converting farmland into solar
22 farms is tempting and controversial at the same time.

23 Agrivoltaic (AV) systems combine crops and PV modules on the same
24 land and aim to resolve this conflict (Dupraz et al., 2011; Dinesh and Pearce,
25 2016; Weselek et al., 2019). By sharing sunlight between modules and plants,
26 a hectare can deliver more food plus energy than the two uses side by side,
27 with land-equivalent ratios (LER) well above one in field trials and mod-
28 elling studies (Dupraz et al., 2011; Amaducci et al., 2018; Valle et al., 2017;
29 Trommsdorff et al., 2021). The partial shade also changes the crop micro-
30 climate. It lowers canopy temperature and evaporative demand, conserves
31 soil moisture and improves water-use efficiency, especially in drylands (Mar-
32 rou et al., 2013b; Hassanpour Adeh et al., 2018; Barron-Gafford et al., 2019;
33 Elamri et al., 2018; Weselek et al., 2021). Agrivoltaics is therefore a can-
34 didate adaptation measure against heat and drought stress, as reviewed by
35 Mamun et al. (2022), in addition to a mitigation measure through renew-
36 able electricity. The same shade, however, also reduces the light available for
37 photosynthesis. Regulators increasingly require AV systems to retain most
38 of the reference crop yield. The German pre-standard DIN SPEC 91434, for
39 example, requires at least 66% of the reference yield (DIN, 2021), and other

schemes and many farmers set stricter floors. How much electricity can be produced without sacrificing food security depends on the crop, the local climate, the year and the growth stage.

Most AV designs fix the geometry of the array, i.e. the ground coverage ratio (GCR, module area per ground area) and the tilt or tracking rule, from generic rules of thumb. Single-axis trackers add a degree of freedom: the modules can be rotated away from the sun to let light through when the crop needs it, as demonstrated for lettuce by Valle et al. (2017), or towards the sun to maximise electricity. Choosing the rotation every day is a sequential decision problem. The value of light depends on the crop’s phenological stage, soil water and upcoming weather, and the effect of a decision appears in the harvest months later. Recent work in this journal has used evolutionary multi-objective optimisation to lay out PV arrays on farmland (Zhang et al., 2026). Machine learning for crop management has mainly used reinforcement learning with black-box simulators (Gautron et al., 2022), which would need very many simulated seasons for global, multi-decadal assessments.

Two methodological strands are relevant. Differentiable process models, which couple physical simulators with learning, allow parameters and policies to be fitted by gradient descent through the simulator (Reichstein et al., 2019; Shen et al., 2023), and we use them for both calibration and control. Decision making under deep uncertainty asks for designs that perform acceptably across plausible models and not only under the best estimate (Marchau et al., 2019), and scenario-based designs have finite-sample guarantees under convexity assumptions that we do not meet (Calafiore and Campi, 2006; Campi and Garatti, 2008); our design rules are heuristic counterparts and we compare them with a simple safety margin.

Two questions follow. First, how much of the electricity that an agri-voltaic system can produce under a yield-retention floor comes from the array density, and how much from dynamic light-sharing control of the trackers? Second, how robust is that answer to the weakest part of any such study, the crop response to shade, and does a controller learned from a simulator add anything to a well-designed hand-written rule? Answering them requires a simulator that couples the physics of light, PV, microclimate and crop growth, a protocol in which all controllers face the same design constraints, a check of the shade response against field data, and enough sites and seasons to separate real effects from noise.

This paper makes three contributions. The first is a quantification of light-sharing control under a yield floor. Light sharing is parameterised by

one daily level between full backtracking and the rotation that minimises the shadow on the crop, and we compare static arrays, a seasonal schedule, a rule that follows crop phenology, a rule that reacts to heat and drought stress, and a neural controller (SALS) trained through the simulator. The second is a deployable evaluation protocol. The array density of every site is chosen from its historical seasons only, with the same rule for every method, and performance is measured on held-out sites and years, over three spatial splits (and, on the main split, three training seeds). The third is a differentiable agrivoltaic simulator that supports both, coupling hourly solar geometry, single-axis trackers with backtracking, row-to-row shading and PV electricity, the direct and diffuse light reaching the crop, canopy temperature and reference evapotranspiration under the modules, and the SIMPLE crop model (Zhao et al., 2019) extended with a flowering-stage shade penalty. All components are differentiable tensor operations (Karniadakis et al., 2021). The PV part is benchmarked against PVGIS. The crop response to shade is learned from the open data of a meta-analysis (Laub et al., 2022), checked against field trials on rice, wheat, potato and maize (Gonocruz et al., 2021; Weselek et al., 2021; Ramos-Fuentes et al., 2023), and propagated through the simulator as an ensemble.

We apply the protocol to 400 crop–site pairs that represent where maize, wheat, rice, soybean and potato are grown worldwide, sampled in proportion to their harvested area (Yu et al., 2020) with local crop calendars (Jägermeyr et al., 2021), and driven by NASA POWER daily weather for 2001–2019 (Sparks, 2018) under baseline and +1.5, +2 and +3 °C climates. The main findings are as follows.

1. Light sharing is the main lever on electricity at a given floor, although economically it pays only at high prices (item 6). At the same mean yield retention of 90%, arrays with light-sharing control carry 2.1 to 2.9 times the module area of static arrays and produce 1.8–2.4 times as much electricity (Section 3.1).
2. The best hand-written rule follows crop phenology and needs no weather forecast. Rules that react to heat or drought stress are not better and depend on forecast quality (Section 3.3).
3. The crop response to shade can be learned from data. A hierarchical model fitted to 428 observations from 58 field studies predicts a held-out study of the four staple crop types to about 0.22 in relative yield, no better than proportional loss (it is better only for crop types the

- 115 simulator does not represent), so between-study spread is large. The
 116 simulator is close to these curves for cereals and potato but milder
 117 than the meta-analysis for maize and soybean, whereas one maize trial
 118 is milder still; the gains for those two crops are therefore uncertain and
 119 possibly overestimated (Section 3.7).
- 120 4. A learned controller adds a small gain over the best rule: 0.044 LER
 121 over the written phenology rule but only 0.007 over a window optimised
 122 on the training seasons of each split. Training it on ensembles of shade
 123 responses, or giving it a farm-measurable biomass signal, does not help,
 124 and with the wider ensembles it did not retain its margin over the
 125 written rule in our training budget (Section 3.8).
 - 126 5. The bottleneck is the uncertainty of the crop response that enters the
 127 design. A density chosen under the mean response leaves about 70% of
 128 crop-site pairs at the floor when another plausible response holds (ora-
 129 cle 85–87%). A rule built on the quantiles of an ensemble of responses
 130 raises this by 8 points for 0.03 LER, and a floor raised by 0.02 under
 131 the mean response gives no detectable difference at equal compliance
 132 (the intervals are too wide to establish equivalence); no rule is safe out-
 133 side the ensemble (whose spread is narrower than the between-study
 134 spread), so the ensemble is useful to size a margin (Section 3.8).
 - 135 6. Under illustrative costs, light sharing breaks even with the open field,
 136 at floor-designed densities, at 70 to 92 USD MWh⁻¹ (static arrays at
 137 59), learning is worth at most about 0.3–0.4 kUSD ha⁻¹ yr⁻¹ over an
 138 optimised window at 90 USD MWh⁻¹ (the 95% intervals reach from
 139 about -0.1 to 0.8, and sensing and compute costs are excluded), and
 140 a conventional plant replacing the crop is more profitable above its
 141 break-even of 48 USD MWh⁻¹ (Section 3.9).
 - 142 7. At the yield floor, agrivoltaics on 5% of the world’s staple cropland
 143 could supply about 14 PWh yr⁻¹, but only with about 10 TWp of mod-
 144 ules, several times the installed PV capacity of the world (Section 3.11).

145 2. Materials and methods

146 2.1. Global cropland sites, weather and crop calendars

147 We studied five staple crops that together cover about 670 Mha of har-
 148 vested area and 580 Mha of physical cropland: wheat, rice, maize, soybean
 149 and potato. Harvested areas (rainfed and irrigated) were taken from the

150 SPAM 2010 gridded production maps (Yu et al., 2020) and aggregated to 1°
 151 cells. For each crop, 80 cells with at least 5000 ha of the crop were drawn
 152 without replacement with probability proportional to harvested area. An un-
 153 weighted average over the sample therefore approximates an area-weighted
 154 average over global production. The 400 crop-site pairs lie in 381 distinct
 155 cells in 50 countries (Fig. 2). Sowing and maturity dates came from the
 156 GGCM Phase 3 crop calendar at 0.5° (Jägermeyr et al., 2021). For wheat
 157 we used the winter-wheat calendar unless the coldest-month mean temper-
 158 ature was below -8°C , in which case the spring-wheat calendar was used.
 159 Daily weather for 2001–2020 was retrieved from NASA POWER (Sparks,
 160 2018): global and diffuse shortwave irradiance, maximum, minimum and
 161 dew-point temperature, precipitation and wind speed at 2 m. Each site con-
 162 tributes 19 seasons (sowing years 2001–2019), each simulated as the 365 days
 163 after sowing, which gives 7600 site-seasons.

164 Climate change was represented by the global warming levels +1.5, +2
 165 and +3 °C relative to 2001–2020. Daily maximum, minimum and dew-point
 166 temperatures were shifted uniformly, which keeps relative humidity approx-
 167 imately constant and raises the vapour-pressure deficit. The atmospheric
 168 CO₂ concentration was set to 390, 450, 500 and 600 ppm for the baseline and
 169 the three warming levels. Precipitation and irradiance were left unchanged.
 170 This delta approach isolates the thermal signal of warming; its limitations
 171 are discussed in Section 4.5.

172 *2.2. A differentiable agrivoltaic simulator*

173 The simulator advances all site-seasons in parallel with a daily time step
 174 for the crop and an hourly time step for radiation and PV (Fig. 1). All equa-
 175 tions are implemented as PyTorch tensor operations (Paszke et al., 2019), so
 176 that yields and electricity are differentiable with respect to the design and
 177 control variables.

178 *Hourly irradiance.* Daily global and diffuse irradiance were disaggregated
 179 into hourly values with the Collares-Pereira and Rabl (1979) and Liu and
 180 Jordan (1960) ratios and renormalised to the daily totals. Beam irradiance
 181 on the horizontal is the difference between the two. Solar position follows
 182 the standard declination and hour-angle equations.

183 *Array geometry and PV electricity.* We consider elevated horizontal single-
 184 axis trackers with north–south axes. The axis height is 1.25 module widths,

185 e.g. 2.75 m for the common 2.2-m single-module-high (1P) tracker. The ro-
 186 tation ϕ is limited to $\pm 60^\circ$. The ground coverage ratio g is the module width
 187 divided by the row pitch and is restricted to $g \leq 0.5$. For 2.2-m modules this
 188 keeps a pitch of at least 4.4 m, and a free strip of at least 2.2 m between the
 189 rows when the modules are horizontal, which is the stow position during field
 190 operations. Let ψ be the solar zenith angle projected onto the east–west ver-
 191 tical plane. The energy-optimal rotation with backtracking (Lorenzo et al.,
 192 2011) is $\phi_{\text{bt}} = \psi - \text{sgn}(\psi) \arccos(\cos \psi / g)$ when $\cos \psi < g$, and $\phi_{\text{bt}} = \psi$ oth-
 193 erwise. The *light-sharing* rotation ϕ_{sh} is the feasible angle closest to edge-on
 194 to the sun, which minimises the shadow on the crop. Given a light-sharing
 195 level $u \in [0, 1]$ chosen once per day, the rotation in every hour is

$$\phi = (1 - u) \phi_{\text{bt}} + u \phi_{\text{sh}}. \quad (1)$$

196 Plane-of-array irradiance combines three terms: beam irradiance on the mod-
 197 ule, reduced by the fraction of the module shaded by the neighbouring row,
 198 $\max\{0, 1 - \cos \psi / (g |\cos(\phi - \psi)|)\}$; isotropic sky diffuse; and ground-reflected
 199 light (albedo 0.2). The cell temperature follows the NOCT model (45°C),
 200 with an hourly air temperature interpolated sinusoidally between the daily
 201 extremes. Electricity per unit land is plane-of-array irradiance $\times \eta \times g$, with
 202 $\eta = 0.21 [1 - 0.0035(T_{\text{cell}} - 25)] (1 - 0.14)$ for module efficiency, temperature
 203 losses and system losses. The installed capacity is $2.1 g \text{ MWp ha}^{-1}$. A con-
 204 ventional ground-mounted PV plant ($g = 0.4$, backtracking, no crop) is the
 205 energy reference E_{PV} .

206 *Validation of the PV model.* We compared the simulated specific yield of an
 207 isolated tracker row ($g = 0.05$, negligible row-to-row shading) with PVGIS 5.2
 208 (European Commission, Joint Research Centre, 2022), for a horizontal north–
 209 south single-axis tracker with 14% losses, at 30 randomly chosen sites (29
 210 returned data). Across sites, the simulated and PVGIS specific yields were
 211 correlated with $r = 0.971$. The simulator was on average 4.9% lower (mean
 212 absolute deviation 5.5%). This is consistent with its $\pm 60^\circ$ rotation limit
 213 and with the different irradiance databases (NASA POWER vs. SARA2,
 214 ERA5 and NSRDB). The benchmark covers an isolated row at low density
 215 only; electricity at high density, with row-to-row shading and diffuse-light
 216 effects, was not validated.

217 *Light and microclimate at the crop.* Averaged over the row pitch, the fraction
 218 of ground reached by beam light is $1 - \min\{1, g |\cos(\phi - \psi)| / \cos \psi\}$. The

fraction of sky diffuse reaching the ground is a two-dimensional isotropic sky-view factor. We pre-computed it for 25 values of g and 13 rotation angles by ray casting over 31 neighbouring rows, and interpolate it bilinearly (differentiably). The daily ratio s of the irradiance at the crop to that of the open field defines the crop’s shading. The canopy maximum temperature is lowered by $\kappa(1 - s)$.

κ is uncertain. Field studies under PV modules report small reductions of daytime crop or canopy temperature and little change in air temperature (Marrou et al., 2013b; Barron-Gafford et al., 2019), and heat-stress thresholds calibrated on air temperature do not transfer directly to canopy temperature (Siebert et al., 2014; Webber et al., 2018). We therefore use a moderate $\kappa = 1^\circ\text{C}$ as the main case and report $\kappa = 0$ (no cooling) and $\kappa = 2^\circ\text{C}$ for all methods (Section 3.2). Nocturnal warming under the modules is not represented. The reference evapotranspiration under the modules, ET_0 , is re-computed with the FAO-56 Penman–Monteith equation (Allen et al., 1998) from the reduced radiation and canopy temperature. Soil water conservation under the modules (Hassanpour Adeh et al., 2018; Elamri et al., 2018) therefore emerges from the model, but rain redistribution by the modules (Marrou et al., 2013a) is ignored.

Crop growth. Crop growth follows the SIMPLE model (Zhao et al., 2019) with the species parameters of its Table 1a (Table 1). Daily biomass growth is

$$\Delta B = R_{\text{crop}} \cdot \text{RUE} \cdot f_{\text{Solar}} \cdot f_{\text{CO}_2} \cdot f_{\text{Temp}} \cdot \min\{f_{\text{Heat}}, f_{\text{Water}}\}, \quad (2)$$

where $R_{\text{crop}} = sR$ is the radiation received by the crop. The terms are as follows.

- f_{Solar} is the canopy’s radiation interception. In SIMPLE it depends only on thermal time, with senescence accelerated by heat and drought stress. Shade therefore reduces the daily biomass increment but not leaf-area development.
- f_{Heat} decreases linearly between the heat threshold T_{heat} and the extreme temperature T_{ext} of the canopy maximum temperature.
- $f_{\text{Water}} = 1 - S_{\text{water}} \text{ARID}$, where the agricultural reference index for drought (Woli et al., 2012) is computed from a daily soil-water balance with curve-number runoff, drainage and root water uptake. Irrigated fields have $f_{\text{Water}} = 1$.

253 SIMPLE has a fixed harvest index and therefore ignores the high sen-
 254 sitivity of grain number to shading around flowering (Fischer, 1985; An-
 255 drade et al., 1999). It also ignores the differences in shade tolerance be-
 256 tween crop types. In the meta-analysis of Laub et al. (2022), maize and
 257 grain legumes lost yield strongly even at low shade levels, whereas C3 ce-
 258 reals and tuber crops initially lost less than proportionally; shading trials
 259 also show marked effects of shade on maize development (Ramos-Fuentes
 260 et al., 2023). We added a stage-dependent penalty. During a critical window
 261 ($0.45 \leq \text{TT}/T_{\text{sum}} \leq 0.65$), the harvest index is multiplied by $1 - h\bar{d}$, where $\bar{d}$
 262 is the mean relative radiation deficit $1 - s$ in that window. The sensitivity h
 263 is highest for maize (0.6) and soybean (0.5), the crop types that Laub et al.
 264 (2022) found most shade-sensitive, lower for the C3 cereals (wheat 0.4; rice
 265 0.5, for which no specific curve was available) and lowest for potato (0.2).
 266 These values are assumptions that reproduce this ranking; they are not cal-
 267 ibrated estimates. Section 3.6 shows the resulting shade-response curves.
 268 Because RUE is linear, SIMPLE does not credit the higher light-use effi-
 269 ciency of shaded canopies under more diffuse light. The penalty, conversely,
 270 makes shading around flowering more damaging than proportional. The net
 271 effect is closer to the field evidence than either assumption alone.

272 The thermal-time requirement to maturity was calibrated for every site as
 273 the mean thermal time between the GGCM sowing and maturity dates under
 274 the 2001–2020 climate, and the canopy parameters were scaled accordingly.
 275 The cultivar is thus adapted to the present climate, and warming shortens the
 276 season. Growth stops smoothly once the thermal-time requirement is met.
 277 Yield is biomass multiplied by the (shade-adjusted) harvest index. Each site
 278 is represented by a rainfed and an irrigated sub-field in proportion to its
 279 SPAM irrigated fraction. Each sub-field carries its own array, so both yield
 280 and electricity are area-weighted over the two sub-fields.

281 2.3. SALS: simulation-trained adaptive light sharing

282 SALS learns a *control network* $u_t = \pi_{\omega}(\mathbf{x}_t; g)$ (Algorithm 1). It is a
 283 multilayer perceptron with two hidden layers of 64 units and a bounded-logit
 284 output, $u = \sigma(5 \tanh(z/5)) \in [0.007, 0.993]$, which keeps gradients non-zero
 285 at the extremes. It chooses the daily light-sharing level of Eq. (1). Its inputs
 286 are the information available on day t :

- 287 • the crop and soil state: relative thermal time TT/T_{sum} , radiation in-
 288 terception f_{Solar} , relative soil water and the previous day’s ARID;

- a *forecast* of the day’s maximum temperature (relative to T_{heat} and T_{opt}), irradiance and ET_0 ;
- static information: the irrigation status of the sub-field, the array’s GCR g , latitude, day of season and the crop.

Forecasts are perturbed with random errors, independently every day: Gaussian with a standard deviation of 2°C for maximum temperature, and multiplicative with standard deviations of 20% for irradiance and 15% for ET_0 . The physics always uses the true weather. The same forecast errors are applied to the rule-based controllers. The controller is conditioned on g . During training, each season is simulated with a GCR close to the density that is sustainable at its training site. This is the largest value on a coarse grid $(0.05, 0.10, \dots, 0.50)$ at which seasonal sharing keeps the mean retention of the site’s training seasons at or above ρ , plus a uniform perturbation in $[-0.05, +0.10]$ truncated to $[0.05, 0.50]$. The network therefore learns to control the densities that design selection will later pick. Outside the cropping season the trackers backtrack for maximum electricity.

For site-season i with open-field yield Y_i^{open} , let $r_i = Y_i/Y_i^{\text{open}}$ be the yield retention and $e_i = E_i/E_{\text{PV},i}$ the electricity relative to a conventional PV plant. Training minimises the penalised objective

$$\mathcal{L}(\omega) = \mathbb{E}_{i \sim \mathcal{B}} \left[-e_i + \mu_1 \max\{0, \rho - r_i\} + \mu_2 \max\{0, \rho - r_i\}^2 \right], \quad \rho = 0.9, \mu_1 = 5, \mu_2 = 30, \quad (3)$$

which combines an exact (hinge) and a quadratic penalty for the yield-retention floor ρ . Because r_i and e_i are outputs of the differentiable simulator, the exact gradient of Eq. (3) is obtained by back-propagation through the 365 daily steps of every season (back-propagation through time). This is the key difference from reinforcement learning with black-box simulators (Gautron et al., 2022), which estimates gradients from sampled returns.

Each minibatch contains 512 site-seasons, each simulated as a rainfed and an irrigated sub-field, drawn from one of the four climates in turn. Seasons with a failed open-field crop ($Y^{\text{open}} < 0.2 \text{ t ha}^{-1}$) are excluded from training and evaluation. The network is fine-tuned through the simulator with Adam (Kingma and Ba, 2015) (learning rate 10^{-3} , cosine decay, gradient-norm clipping at 1) for 300 steps. Growth-stage logistic functions are evaluated in a numerically stable form, so that gradients remain finite even under extreme heat.

Algorithm 1 SALS training

Require: training site-seasons in four climates; floor ρ ; penalties μ_1, μ_2

- 1: initialise π_ω by behaviour cloning of the phenology rule
- 2: **for** step = $1, \dots, N$ **do**
- 3: sample a climate and a minibatch $\mathcal{B}$ of site-seasons (rainfed and irrigated sub-fields)
- 4: set g_i to the site’s sustainable density plus a perturbation; initialise crop and soil states
- 5: **for** day $t = 1, \dots, 365$ **do**
- 6: $u_{it} \leftarrow \pi_\omega(\mathbf{x}_{it}; g_i)$ from state and noisy forecast if the crop is in the field, else 0
- 7: hourly rotation (Eq. 1), PV electricity, crop light, canopy temperature, ET_0
- 8: advance the soil-water balance and the SIMPLE crop state (Eq. 2)
- 9: $r_i \leftarrow Y_i/Y_i^{\text{open}}, e_i \leftarrow E_i/E_{\text{PV},i}$ (area-weighted); loss $\mathcal{L}$ (Eq. 3)
- 10: back-propagate through the season; Adam update of ω

322 *Initialisation.* Trained from random weights, the controller converged to
323 sharing light throughout the cropping window ($u > 0.99$ on all days), i.e. to
324 the seasonal-sharing rule, because the gradient of electricity with respect to u
325 is small and of the same sign for most days of a season. We therefore initialise
326 the network by behaviour cloning of the phenology rule of Section 2.4: binary
327 cross-entropy against the rule’s output (soft targets 0.95 and 0.05) on about
328 1.7 million day-states visited by the rule in the training seasons, with 1500
329 Adam steps at a learning rate of 3×10^{-3} . Fine-tuning then moves the
330 policy away from this initialisation wherever doing so improves the objective.
331 Consequently, the gain of SALS over the phenology rule measures what fine-
332 tuning through the simulator adds to that rule, not what can be learned
333 without it.

334 *2.4. Deployable design selection and baselines*

335 A real array has one fixed density, which must be chosen before it is built.
336 Every method, SALS included, therefore uses the same *history-based design*
337 *rule*. For a site, the GCR is the largest value on the grid $\{0.05, 0.075, \dots, 0.50\}$
338 for which the method’s mean yield retention over the site’s 2001–2014 sea-
339 sons is at least $\rho = 0.9$. If no value qualifies, no array is built and the site
340 remains open field. A stricter *chance-constrained* variant requires $r \geq 0.9$ in

at least 80% of the historical seasons. The historical seasons of the test sites are never used for training, and the weather of the evaluation years is never used for design. We compare four controllers:

- *AV static*: backtracking trackers without light sharing ($u = 0$);
- *Seasonal sharing*: the light-sharing rotation ($u = 1$) throughout the cropping season;
- *Stress rule*: an expert rule that shares light ($u = 1$) during the main growth phase ($0.1 < TT/T_{\text{sum}} < 0.9$) except on days with a heat-stress forecast ($T_{\text{max}} \geq T_{\text{heat}}$) or drought ($\text{ARID} > 0.5$), when the modules track the sun and shade the crop;
- *SALS*: the learned controller.

The open field and a conventional PV plant without crops ($\text{LER} = 1$ each) are the references. Because the design rule is identical for all controllers, differences between methods reflect the quality of control and the array density that each controller can sustain under the yield-retention floor. We also compare the controllers at identical GCR over the whole grid (matched-density frontiers).

2.5. A shade-response model learned from field data

The simulator needs a crop response to shade that is supported by observations. We learned one from the open data of the meta-analysis of Laub et al. (2022) (Zenodo 10.5281/zenodo.5716091, CC BY 4.0): 428 observations of yield relative to an unshaded control at relative shade reductions of 0–90% from 58 studies, with crop and crop type, experiment type (artificial shading or intercropping), climate zone, irradiation and shade type. Relative yield is modelled as $y(s) = \exp(-\beta s^\gamma)$ with Student- t errors ($\nu = 4$; maximum a posteriori estimation, so this is a hierarchical shrinkage regression with bootstrap intervals and not a full Bayesian model). We compared six models: proportional loss $y = 1 - s$; one curve for all crops; one curve per crop type without pooling; one curve per crop type with partial pooling (Gaussian shrinkage of $\log \beta$ and $\log \gamma$ towards a common value, scale 0.5); partial pooling with covariates on $\log \beta$ (standardised irradiation, subtropical zone, solar-panel shade and intercropping); and gradient-boosted trees on the same inputs. Fits use L-BFGS. Models were compared by leave-one-study-out cross-validation, in which every study is held out once, which tests

375 the prediction of a new study and not of a new observation in a known study.
 376 Intervals for the crop-type curves and covariate effects come from a cluster
 377 bootstrap over studies (150 resamples).

378 2.6. *Shade-response uncertainty: gradient calibration, robust controllers and* 379 *design*

380 The crop’s response to shade is the least certain part of the simulator,
 381 so we treat it as an unknown with a distribution. Two parameters per crop
 382 enter the simulator: a radiation-use compensation c (growth is multiplied by
 383 $1 + c(1 - s)$, s being the share of open-field irradiance that reaches the crop)
 384 and a multiplier h on the flowering-window harvest-index penalty. Because
 385 the simulator is differentiable, both can be fitted by gradient descent through
 386 whole simulated seasons to published yield–shading observations. The ob-
 387 servations are the relative yields at 20% and 40% shading of the data-driven
 388 curves of Section 2.5 for the five crops (median, with an error equal to the
 389 width of the 80% bootstrap interval divided by 2.56 and at least 0.04; rice
 390 takes the curve of C3 cereals, which contains no rice), the field trials on rice,
 391 wheat, potato and maize (Gonocruz et al., 2021; Weselek et al., 2021; Ramos-
 392 Fuentes et al., 2023) (each season of each trial as a separate observation, error
 393 0.06–0.08) and the unaffected harvest index of shaded maize (Ramos-Fuentes
 394 et al., 2023). We call the resulting ensemble E2. A wider ensemble, E3, uses
 395 the same procedure with a target error of 0.2 for the learned curves, which
 396 is the between-study spread of a new study (Section 3.7); the spread of
 397 the resulting members is smaller (standard deviation of relative yield across
 398 members 0.04–0.09 at 20% shading and 0.05–0.12 at 40% shading, against
 399 0.02–0.04 and 0.03–0.06 for E2), because the fit also weights the field trials
 400 and the priors. An earlier ensemble, E1, was fitted in the same way to curves
 401 read by hand from the figures of Laub et al. (2022) and was used to train
 402 the robust and adaptive controllers below. Priors are $c \sim \mathcal{N}(0.4, 0.5^2)$ and
 403 $h \sim \mathcal{N}(1, 0.5^2)$, and the parameters are clamped to $[0, 2.5]$ (c) and $[0, 2]$ (h)
 404 after every step, which is why a response milder than the data cannot be
 405 lowered further. We fitted 12 ensemble members by randomised maximum a
 406 posteriori estimation: every member sees the observations perturbed by their
 407 errors and a prior draw perturbed by its width, and is optimised with Adam
 408 (40 steps, with the loss still falling slightly at the last step; 40 training-block
 409 seasons per crop and member; shading of the static arrays cached). The
 410 ensemble is a device for generating plausible responses and not a validated
 411 posterior: it has no coverage check, and for some parameters (the harvest-

index multiplier of wheat and potato) its spread equals that of the prior. The design analyses use five fixed members (indices 0, 3, 5, 8 and 10 of 12), chosen in advance. The ensemble is used in three ways. All training and evaluation ran on a 4-core CPU; one training run takes about 20–25 minutes and the calibration of one ensemble about 7 minutes.

Tuned rules. The written phenology rule shares light while thermal time lies between 0.05 and 1.0 of the season requirement. To avoid comparing a trained controller with an untuned rule, we tuned this window on the training-block seasons of each split (500 random seasons, baseline climate, main shade response) with the objective used for all rules: the relative electricity at the density where the mean retention equals the floor, found by interpolation over five densities (0.1 to 0.5). A 4×3 grid of start (0, 0.05, 0.15, 0.3) and end (0.6, 0.8, 1.0) gave the grid-tuned window. A five-parameter schedule (start in $[0, 0.6]$, end in $[0.5, 1.2]$, two ramp widths and a level in $[0, 1]$, all through logistic or exponential transforms) was then optimised by CMA-ES (Hansen and Ostermeier, 2001) with 300 seasons, population 8 and 25 generations (200 evaluations) per run; run 1 started at the grid-tuned window and runs 2 and 3 at random points, and the run with the best training objective was evaluated on that split (the CMA-ES schedule and the grid window were tuned separately on each split’s training block; on split 0 run 2 was best, on split 1 run 2 and on split 2 run 1). The grid window was 0.15 to 1.0 on all three splits, and the schedules of the best runs had starts of 0.20, 0.20 and 0.22, ends of 0.93, 0.92 and 0.93, a level of 1.0 and ramp widths below 0.01 of the requirement. On split 0 run 3 stopped in a worse optimum (objective 0.345 against 0.377) and on split 1 as well (0.349 against 0.379), so three runs give no guarantee of global optimality.

Robust and adaptive controllers. SALS-robust has the architecture and training of SALS (Section 2.3), but every training season is simulated with the response of a randomly drawn ensemble member that the controller does not observe, and the density jitter is widened to $[-0.05, +0.25]$ because feasible densities differ between members. SALS-adaptive additionally observes the realised biomass of the crop under the modules relative to an open reference plot, a quantity that can be measured on the farm by remote sensing or by a reference plot. A feedback rule that shares light whenever this ratio falls below a target serves as the hand-written counterpart; its target (1.0) was chosen on training-block seasons under the ensemble.

Uncertainty-aware design. The design rule of Section 2.4 chooses the largest GCR whose historical retention meets the floor. Under shade-response

450 uncertainty we choose the largest GCR that meets the floor in at least 75%
 451 of the ensemble members (three of four in the leave-one-out analysis, four
 452 of five otherwise), and compare it with a design under the ensemble mean
 453 and with an oracle design under the true response. Evaluation of this rule
 454 is leave-one-out: the truth is the held-out member and the design uses the
 455 other four.

456 *Design rules compared.* With the ensemble members indexed k and one
 457 member held out as the truth, the density of a pair is chosen from its 2001–
 458 2014 seasons as (i) the *oracle*: the largest GCR whose mean retention under
 459 the truth meets the floor; (ii) the *mean-response design*: the same rule under
 460 the mean response of the other four members (the mean of their parameters);
 461 (iii) a *margin*: the same rule under that mean response with the floor raised
 462 to 0.92, 0.94 or 0.96; and (iv) the *ensemble rule*: the largest GCR at which
 463 the mean retention under at least a share q of the other four members meets
 464 the floor ($q = 0.5$: two of four; 0.75 : three of four; all: four of four). Out
 465 of the ensemble, the five members and their mean (the mean of all twelve
 466 members for the mean-response design) are used in the same way and the
 467 truth is one of the four fixed responses below. The margin grid was chosen
 468 after a first analysis of one ensemble and is not pre-specified.

469 *Stress tests.* Controllers and rules are also evaluated under four fixed
 470 responses, the proportional response, the response calibrated to the meta-
 471 analysis, the response fitted to the field trials and a harsh response (propor-
 472 tional growth and a 1.5 times larger harvest-index penalty). The proportional
 473 and harsh responses lie outside the range of the ensemble.

474 2.7. Economic analysis

475 To judge whether the electricity gains matter economically, we computed
 476 an annual net value per hectare for each method relative to the open field:
 477 electricity revenue minus the annualised capital cost (annuity) and operating
 478 cost of the installed capacity minus the loss of crop revenue (crop produc-
 479 tion costs are unchanged). The installed capacity is 2.1 g MWp per hectare
 480 (modules cover a share g of the land at 0.21 kWp m^{-2}). Yield is dry matter,
 481 valued at assumed prices. A conventional ground-mounted plant (GCR 0.4)
 482 that replaces the crop is the comparator. All parameters (Table 15) are as-
 483 sumptions chosen to be representative, not data, and their ranges drive the
 484 conclusions; we therefore report break-even electricity prices and one-at-a-
 485 time sensitivities, and we do not monetise food security, which market crop
 486 prices do not capture. The analysis omits land rent, module degradation,

the extra structure cost of sparse arrays, curtailment, subsidies and the uncertainty of the parameters themselves, which the bootstrap intervals do not include.

2.8. Evaluation protocol, statistics and upscaling

Sites were split spatially: $10^\circ \times 10^\circ$ blocks were randomly assigned to training (70%) or test (30%), so that no test site has a training site in the same block. Training uses the 2001–2014 seasons of the training sites. Design selection uses the 2001–2014 seasons of the test sites. Evaluation uses the 2015–2019 seasons of the test sites. The same site-seasons are analysed in all four climates.

We report the following metrics:

- the mean electricity ($\text{MWh ha}^{-1} \text{yr}^{-1}$) and the relative electricity e ;
- the mean yield retention r and the production-weighted retention $\sum Y / \sum Y^{\text{open}}$;
- season-level compliance (share of seasons with $r \geq \rho$) and site-level compliance (share of sites whose mean test-season retention is $\geq \rho$);
- the lower tail (10th percentile of r , share of seasons with $r < 0.8$);
- the land-equivalent ratio $\text{LER} = r + e$ (Dupraz et al., 2011), which gives equal weight to the retained yield and to the electricity relative to a conventional plant, and is relative to the open field of the same climate (a higher LER under warming reflects the lower reference electricity share and not a benefit of agrivoltaics).

Two aggregations are used and labelled in the text: tables report means over all test seasons of a sample (season-pooled), whereas statistical comparisons and intervals use the crop–site pair as the unit of analysis (mean of the pair’s seasons, then mean over pairs), so the same contrast can differ slightly between the two (for example 0.034 and 0.038 for SALS minus the phenology rule on the main split). Samples are also labelled: the main analysis uses the 92 held-out pairs of one split and a GCR grid with step 0.025; analyses pooled over three splits use the 76 distinct pairs of every third held-out pair and a grid with step 0.05, and analyses of alternative shade responses on one split use every sixth (21 pairs) or every fourth pair (22 pairs). Statistical comparisons use site means as the unit of analysis. We report paired

Table 1: SIMPLE crop parameters (Zhao et al., 2019, Table 1a) and the harvest-index shade sensitivity h added in this study. T_{sum} , I_{50A} and I_{50B} are rescaled per site to the local crop calendar (Section 2.2). Units: temperatures in $^{\circ}\text{C}$, thermal times in $^{\circ}\text{C d}$, RUE in gMJ^{-1} ; S_{CO_2} is the relative RUE increase per 100 ppm above 350 ppm.

Crop (cultivar)	h	T_{sum}	HI	I_{50A}	I_{50B}	T_{base}	T_{opt}	RUE	$I_{50 \text{ max } H}$	$I_{50 \text{ max } W}$	T_{heat}	T_{ext}	S_{CO_2}	S_{water}
Wheat (Batten)	0.4	2150	0.34	280	50	0	15	1.24	100	25	34	45	0.08	0.4
Rice (IR72)	0.5	2300	0.47	850	200	9	26	1.24	100	10	34	50	0.08	1.0
Maize (McCurdy 84aa)	0.6	2050	0.50	500	50	8	28	2.10	100	12	34	50	0.01	1.2
Soybean (Williams 82)	0.5	2350	0.40	600	200	6	27	0.86	120	20	36	50	0.07	0.9
Potato (Sebago)	0.2	2400	0.85	500	350	4	22	1.30	50	30	34	45	0.10	0.4

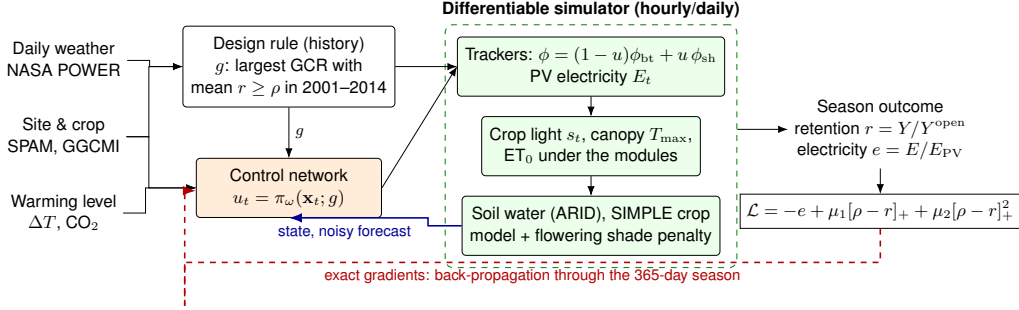


Figure 1: Simulator and controller. The array density g of a site is fixed from its historical seasons. A control network maps the daily crop and soil state, the density g and a noisy forecast to the light-sharing level u_t . The differentiable simulator converts these decisions into PV electricity, light and microclimate at the crop, soil water and crop growth. The season’s electricity and yield retention enter a penalised objective whose exact gradient is back-propagated through the 365-day season.

519 Wilcoxon signed-rank tests and cluster (site) bootstrap 95% confidence in-
520 tervals of the mean difference. SALS was trained with three random seeds
521 on the main split, and with one seed on each of two further spatial splits.

522 For the global estimate, every site is evaluated in the split in which it is
523 held out. The mean electricity and yield per hectare of each crop are scaled
524 to the *physical* area of that crop in SPAM 2010, which avoids double counting
525 multiple-cropped land (Waha et al., 2020). We report the installed capacity
526 (TWp), the capacity factor and a supply curve.

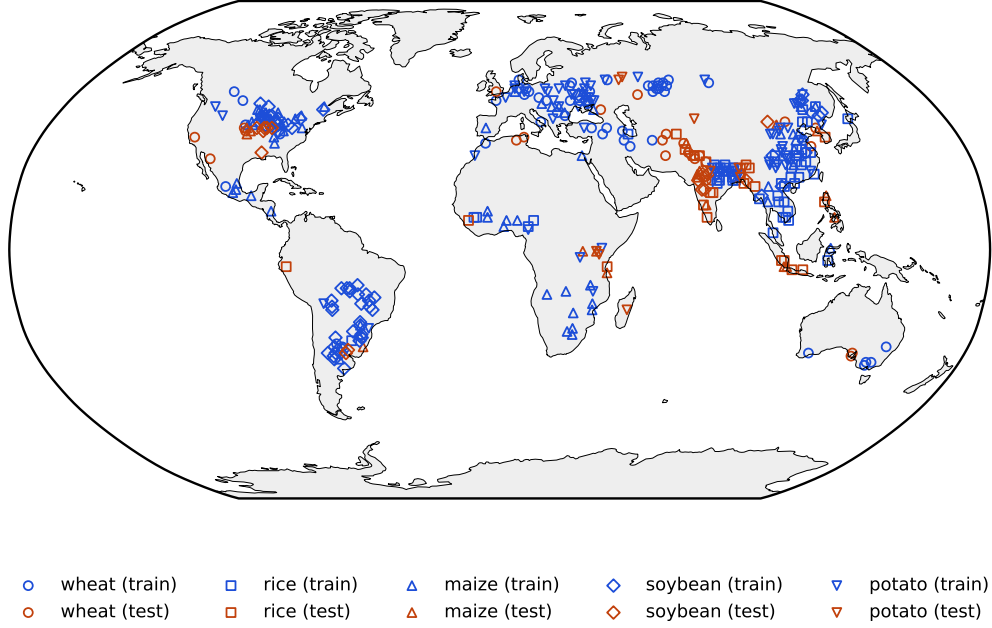


Figure 2: The 400 crop–site pairs, sampled in proportion to the harvested area of each crop (SPAM 2010). Blue: training sites; orange: geographically held-out test sites of the main split.

527 3. Results

528 3.1. Electricity and food on held-out sites and years

529 Table 2 compares the four controllers on the 92 held-out crop–site pairs of
 530 the main split (95 pairs are held out; pairs without a season whose open-field
 531 yield is at least 0.2 t ha^{-1} in all four climates are excluded) (seasons sown
 532 in 2015–2019), after each site’s density has been chosen from its 2001–2014
 533 seasons with the same rule for all controllers. By construction all methods
 534 keep the mean yield retention close to 90% (0.904–0.917), so differences in
 535 electricity are differences in the density that each controller can sustain under
 536 this floor.

537 **Static arrays are the weakest.** Without light sharing, the floor limits
 538 the ground coverage to 6% on average, and arrays produce $217 \text{ MWh ha}^{-1} \text{ yr}^{-1}$,
 539 16% of a conventional PV plant per hectare (LER 1.07). Rotating the
 540 modules to let light through during the cropping season (seasonal shar-
 541 ing) allows a coverage of 17% and $397 \text{ MWh ha}^{-1} \text{ yr}^{-1}$ (LER 1.19). A rule

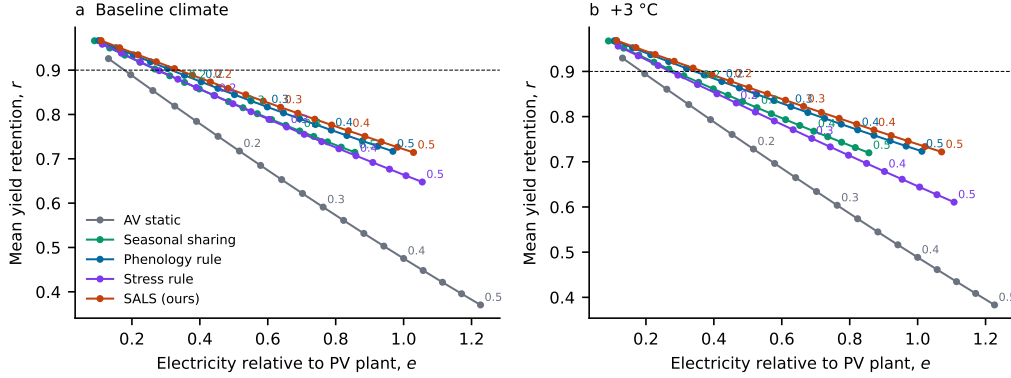


Figure 3: Electricity-retention frontiers at identical array densities (labels: GCR) for the held-out sites. Each point is the mean over test seasons at a fixed GCR; the dashed line is the 90% floor.

that shares light only while the crop is growing (phenology rule) reaches 465 MWh ha⁻¹ yr⁻¹ (LER 1.24). An expert rule that additionally shades the crop on forecast heat or drought days (stress rule) is worse than the phenology rule (LER 1.18): its density is limited to 13% because it reduces light at times when the floor is most binding.

The learned controller is best by a modest, consistent margin. SALS sustains a density of 17% (similar to the phenology rule), produces 510 MWh ha⁻¹ yr⁻¹ and reaches an LER of 1.270 at the same retention. Compared with the phenology rule this is +0.038 in LER (95% CI over sites 0.025–0.053; +10% electricity) and it is higher at 84% of the 92 sites (Table A.18). Compared with static arrays it is +0.205 (0.176–0.236), with 2.35 times the electricity at 2.9 times the density. The frontier in Fig. 3 shows the same picture at matched densities: the learned policy lies on or slightly above the phenology rule over the whole density range, both lie well above seasonal sharing and the stress rule, and static arrays are far below.

What the floor does and does not guarantee. Enforcing the floor on the historical mean does not guarantee it in every season. In the test seasons, 70% of SALS seasons have $r \geq 0.9$ (phenology rule 69%, stress rule 76%, static arrays 93%), and 2.7% have $r < 0.8$ (1.3%, 0.9% and 0.9%). The lower tail of SALS is therefore slightly heavier than that of the rules: its 10th percentile of r is 0.883, against 0.887, 0.886 and 0.903. Requiring $r \geq 0.9$ in at least 80% of a site’s historical seasons (chance-constrained design) raises the season-level compliance to 90–94% for the light-sharing

565 controllers. It costs about 0.065 in LER for SALS (1.205), which remains the
 566 best controller (phenology rule 1.192; Section 3.2).

567 3.2. Robustness

568 **Seeds and splits.** Three training seeds of SALS give LERs of 1.270,
 569 1.268 and 1.269 under the baseline climate and 1.297, 1.298 and 1.297 at
 570 +3 °C. On two further spatial splits, in which different blocks of sites are
 571 held out, SALS exceeds the phenology rule by 0.047 and 0.041 in LER under
 572 the baseline climate and by 0.052 and 0.039 at +3 °C (main split, season-
 573 pooled: 0.034 and 0.032; the site-level paired contrasts of Table A.18 are
 574 0.038 and 0.034). In every split SALS ranks first, the phenology rule second
 575 and static arrays last.

576 **Canopy cooling.** The canopy-cooling coefficient κ is the most un-
 577 certain microclimate parameter. We repeated the whole protocol, includ-
 578 ing the historical design step, with $\kappa = 0, 1$ and 2 °C. Electricity changes
 579 with κ because cooler canopies tolerate denser arrays (SALS: 475, 510 and
 580 549 MWh ha⁻¹ yr⁻¹ under the baseline climate), but the retention is held at
 581 0.905 by the design rule and the ranking is unchanged. SALS exceeds the
 582 phenology rule by 0.029, 0.034 and 0.042 in LER under the baseline climate
 583 (0.022, 0.032 and 0.051 at +3 °C).

584 **Which ingredients matter.** Two ablations test the design of SALS
 585 at the same training budget. Masking the stress inputs (soil water, ARID,
 586 and the maximum temperature, irradiance and ET₀ forecasts) leaves the
 587 LER unchanged (1.269 vs 1.270 under the baseline climate, 1.297 vs 1.297
 588 at +3 °C). Training only on the baseline climate gives 1.268 and 1.291. The
 589 improvement over the rules therefore does not depend on stress-aware inputs
 590 or on training in warmed climates.

591 3.3. Floor conditioning, chance-constrained training and forecast errors

592 **One network for all floors.** Regulations and farmers set different
 593 floors. The network of the main experiments was trained at $\rho = 0.9$ only.
 594 We evaluated it at floors of 0.80, 0.85, 0.90 and 0.95, re-running the historical
 595 design step for each floor (Table 3). Its LER exceeds that of the phenology
 596 rule by 0.037, 0.036, 0.034 and 0.033 under the baseline climate (0.040, 0.037,
 597 0.032 and 0.033 at +3 °C), so the benefit carries over to other floors without
 598 retraining. A network that received the floor as an additional input and was
 599 trained over all four floors did not improve on it (differences in LER ≤ 0.001).

600 Because the design step already adapts the density to the floor, conditioning
 601 the controller on the floor is not needed.

602 **Layout, rotation limit and harvest-index assumptions.** We checked
 603 three assumptions that could drive the ranking (Table 4; 21 pairs of the main
 604 split, one training seed). The rotation limit of the trackers sets how much
 605 light sharing can deliver: with a limit of 45° the LER of the phenology rule
 606 is 1.138, with 60° 1.170 and with 80° 1.206, whereas static trackers stay at
 607 1.05–1.06, so the benefit of light sharing grows with the limit and the ranking
 608 is unchanged. Fixed non-tracking arrays held flat or rotated by 30° about
 609 the north–south axis (facing east, not the equator; equator-facing tilts are
 610 not modelled) do no better than static trackers (LER 1.048 and 1.043 against
 611 1.053; differences of 0.005 are within the noise of 21 pairs and one training
 612 seed), so static trackers are no worse than the fixed arrays we tested; verti-
 613 cal bifacial layouts are not modelled. The ranking of the rules also does not
 614 depend on the assumed harvest-index penalty. The window test is weakly in-
 615 formative about the timing of the phenology rule, which shares light through
 616 almost the whole growing period, so a shifted window cannot move it much;
 617 the informative case is the removal of the penalty. Moving the critical window
 618 to 0.30–0.50 or 0.60–0.80 of the thermal-time requirement changes the LER
 619 of the phenology rule by at most 0.001, and removing the penalty altogether
 620 raises all light-sharing LERs (phenology rule 1.328, SALS 1.353) without
 621 changing the order: the phenology rule exceeds seasonal sharing by 0.03 to
 622 0.05 in every case (0.035 with the early window, 0.034 with the late window,
 623 0.053 without penalty), because it produces electricity before canopy devel-
 624 opment and after maturity, when the crop does not use the light, and not
 625 because of the assumed window. The margin of SALS over the written rule
 626 changes with these assumptions (0.005 to 0.025 LER), and its margin over
 627 the tuned rule is 0.000 to 0.009 (at a rotation limit of 45° SALS equals the
 628 tuned rule).

629 **A floor of 66%.** The German pre-standard DIN SPEC 91434 requires
 630 66% of the reference yield (DIN, 2021). Repeating the design step with
 631 $\rho = 0.66$ (baseline climate, main split, same network trained at 0.9) the
 632 ground coverage cap of 0.5 binds for the sharing methods (mean GCR 0.48
 633 for the phenology rule and 0.48 for SALS, 0.44 for the stress rule), static
 634 arrays reach a coverage of 0.24 and LER 1.28, the phenology rule 1.66 and
 635 SALS 1.71. At this floor the density is limited by the cap and by the mean
 636 retention of 0.73 reached at the cap, not by the floor, so the comparison of
 637 controllers says little about control; we do not interpret it further.

638 **Chance-constrained training.** We also trained a network whose ob-
639 jective penalises a batch share of seasons below the floor that exceeds 20%.
640 It did not improve the tail. With the mean-based design step its LER was
641 1.263 (baseline) and 1.290 (+3 °C) against 1.270 and 1.297, and with chance-
642 constrained design it reached season-level compliance of 92.6% against 92.0%
643 and an LER of 1.198 against 1.205. The season-level guarantee is obtained
644 by the design step (Section 3.1), not by the training objective.

645 **Forecast errors.** We raised the forecast errors seen by the controllers
646 to 0, 1, 2 and 3 times the errors used in training. The LER of SALS and
647 of the phenology rule was identical at all four error levels to three decimals
648 (densities 0.2 and 0.3: SALS 1.307 and 1.457, phenology rule 1.278 and
649 1.416). Neither controller uses the weather forecast: both rely on the crop’s
650 thermal time, which is observable from on-farm temperature records. The
651 stress rule, which depends on the forecast of maximum temperature, degrades
652 with forecast error: at density 0.3 its retention falls from 0.778 to 0.751 and
653 its LER from 1.432 to 1.418 as the errors triple.

654 **Tuned rules.** The phenology rule above was written, not tuned, whereas
655 SALS is trained. We therefore tuned its window on the training-block seasons
656 with the same objective as the other rules (electricity at the density where
657 mean retention equals the floor), first on a coarse 4×3 grid and then, as a
658 derivative-free baseline, by CMA-ES over a five-parameter schedule (start,
659 end, two ramp widths and a level; three runs of 200 evaluations, Methods).
660 The grid-tuned window starts at 0.15 instead of 0.05 of the thermal-time
661 requirement (Table 5). Two of the three CMA-ES runs, one started at the
662 grid-tuned window and one at a random point, reached the same almost
663 binary window (all three runs stopped at the generation cap) from 0.20 to
664 0.93 of the requirement at level 1.00; the third stopped in a worse optimum.
665 Pooled over three splits and 76 distinct crop–site pairs, the grid-tuned win-
666 dow raises LER over the written rule by 0.026 (95% cluster-bootstrap interval
667 0.016 to 0.037), the CMA-ES window by 0.036 (0.023 to 0.052) and SALS by
668 0.044 (0.030 to 0.060), so the optimisation of a window accounts for about
669 84% of the margin of SALS over the written rule. SALS exceeds the CMA-
670 ES window by 0.007 LER (0.002 to 0.013), which is 2.3% more electricity at
671 equal retention, and the grid-tuned window by 0.017 (0.012 to 0.024). The
672 remaining margin is small (it covers the pairs only: SALS has one training
673 seed per split, and the CMA-ES schedule and the grid window were re-tuned
674 on each split’s own training block); it is not explained by the tuning of a
675 window or a ramp, and it cannot come from soil water, forecasts or aridity,

676 because masking them leaves SALS unchanged (Section 3.2); canopy devel-
 677 opment and density remain untested. SALS and the CMA-ES window meet
 678 the floor in 83% of seasons, the written, grid-tuned and seasonal rules in 84–
 679 85%. Per split, SALS exceeds the CMA-ES window by 0.007 (95% interval
 680 0.001 to 0.013; 32 pairs), 0.010 (0.001 to 0.023; 26 pairs) and 0.005 (–0.002
 681 to 0.012; 35 pairs), with per-pair medians of 0.001 to 0.002, so the margin is
 682 positive on every split but concentrated in a few pairs. The design step quan-
 683 tises the density to a grid of 0.05 (about 30% of the mean density), which is
 684 much coarser than the margin. We therefore also compared the controllers
 685 on the test seasons at matched mean retention, reading the LER at a reten-
 686 tion of 0.90 off the retention–LER curve of each pair (interpolated over the
 687 grid): SALS exceeds the CMA-ES window by 0.0035 (95% interval –0.0003
 688 to 0.0073; 73 pairs), the grid-tuned window by 0.011 (0.007 to 0.016) and the
 689 written rule by 0.036 (0.023 to 0.051). At fixed densities of 0.10 to 0.30 the
 690 difference to the CMA-ES window is 0.001 to 0.003 LER, with intervals that
 691 include zero. At matched retention the margin over the optimised window is
 692 therefore not distinguishable from zero.

693 3.4. *What the learned policy does*

694 Averaged over the rainfed test seasons of the main split at a density of
 695 0.3, the light-sharing level of SALS is a smooth function of relative thermal
 696 time. It is close to zero in the first 10% of the thermal time required for
 697 maturity, rises through 0.55 at 10–20%, stays at 0.94–0.99 between 30% and
 698 70%, and falls through 0.58 at 80–90% to about zero at maturity. The plateau
 699 overlaps the window in which our crop model penalises shading through the
 700 harvest index; we did not test whether this alignment is learned or inherited
 701 from the initialisation. The ramps at both ends let the modules produce
 702 electricity when additional light is worth least to the crop. Fig. 4 shows two
 703 example seasons. The phenology rule, which is also the initialisation of the
 704 network, shares light during the whole growing period. Fine-tuning through
 705 the simulator found that sharing can be reduced at the edges of the season:
 706 this pattern describes the average policy at a density of 0.3 (the designed
 707 mean is 0.17) and we did not test that it causes the gain; the CMA-ES
 708 window is almost binary, so a smooth ramp is not needed to reproduce most
 709 of it.

710 The policy also shares less light on days with a forecast heat stress: the
 711 mean level is 0.35 on such days against 0.60 (baseline) and 0.51 (+3 °C)
 712 on other days, and 58–63% of sites show a lower level on heat days. This

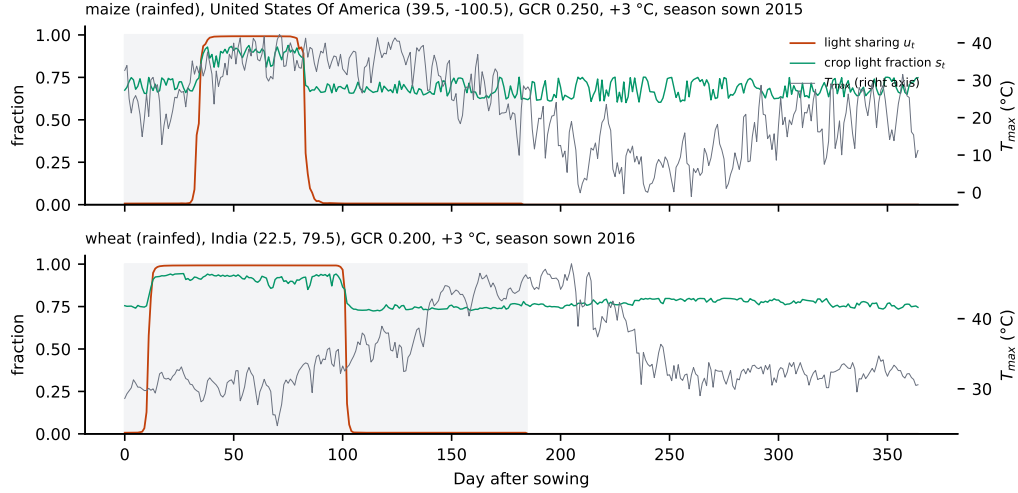


Figure 4: Learned daily control for two test seasons under $+3^{\circ}\text{C}$, at the density chosen for the site, simulated for the dominant sub-field of the site. Shading marks the cropping window. Orange: light-sharing level u_t ; green: fraction of open-field irradiance reaching the crop; grey: daily maximum temperature (right axis).

cannot be read as a stress response, however. Heat days occur late in the season, when the phenological ramp is already falling, and a policy without any stress input shows a similar contrast (0.37 vs 0.57 in a separate check). The ablation above confirms that stress information does not improve the result.

3.5. Crops and regions

The benefit of light sharing differs between crops (baseline climate, SALS). Soybean and maize tolerate the densest arrays (GCR 0.24 and 0.19; 691 and 577 $\text{MWh ha}^{-1} \text{yr}^{-1}$; LER 1.41 and 1.30), potato follows (0.19; 542; 1.31), and rice and wheat are the most restrictive (0.13 and 0.15; 395 and 428; 1.20 for both). The stress rule has a lower LER for every crop (1.26 for soybean, 1.14–1.21 for the others).

The floor is not met evenly. Production-weighted retention is 0.888 for soybean and 0.898 for maize, below 0.9, because high-yielding seasons lose more than low-yielding ones. Regionally, Latin America and Oceania (4 sites) combine the highest density (GCR 0.41) and electricity (1090 $\text{MWh ha}^{-1} \text{yr}^{-1}$) with a production-weighted retention of only 0.862. Africa and the Middle East (8 sites) reach 0.894, and only 63% of their sites keep a mean retention

of at least 0.9. South Asia, the region with the most sites (39), reaches 0.902 (67% of sites) at a density of 0.14. The regional figures rest on 4 to 39 pairs without intervals and should be read as indications. A regulatory floor enforced per site would therefore need a region- and crop-specific design step, not a global one.

3.6. Model behaviour and validation

PV output. The simulated specific yield of an isolated tracker row agrees with PVGIS 5.2 (European Commission, Joint Research Centre, 2022) at 29 sites with $r = 0.971$, a mean bias of -4.9% and a mean absolute deviation of 5.5% (Section 2.2).

Shade response. Fig. 5 shows the simulated relative yield of rainfed crops under static arrays as a function of the season-mean shading of the crop. Yield declines close to proportionally with shading: 84–89% relative yield at 12% shading, 70–78% at 25%, 57–67% at 37% and 35–46% at 57%. Wheat loses most and potato and maize least. This response is steeper than the *less than proportional* losses that Laub et al. (2022) report for small shade levels of C3 cereals and tubers, and it does not reproduce their finding that maize and grain legumes are more shade sensitive than C3 cereals. In the simulator, shading also relieves heat and drought in rainfed maize and soybean, which offsets the higher harvest-index penalty we assigned to them. The absolute effect of shading on each crop is therefore uncertain (Section 4.5).

Comparison with field trials. We fitted one radiation-use compensation coefficient per crop to the yield–shading curves of Laub et al. (2022) using training-block sites only (Methods, calibration script in the repository). The fit changed wheat and rice only (coefficients 0.4 and 0.1) because the proportional model already lies within about 0.02 of the C3-cereal and tuber curves, and it cannot lower the maize and legume responses below the proportional case (root-mean-square error 0.16 for maize and 0.08 for soybean). Against field trials (not independent of the choice of ensemble, Table 6) (Table 6) the model underestimates retained yield: rice 0.79 against 0.84 (Gonocruz et al., 2021), wheat and potato 0.74 and 0.72 against 0.81–0.82 in the worse season of Weselek et al. (2021) (and above 1 in the better one), and maize 0.70–0.75 against 0.83–0.91 at 30–35% shading (Ramos-Fuentes et al., 2023), where harvest index was not affected by shade. The model is therefore conservative for food retention with respect to these trials, except for the stronger maize losses of the meta-analysis. Table 7 shows the consequence for the design results: with the calibrated response the allowed density and the LER of every

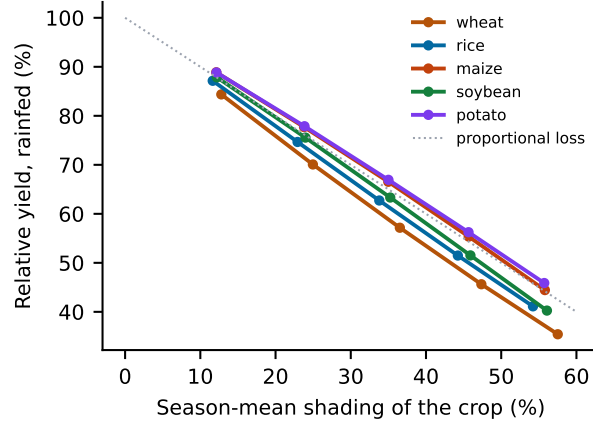


Figure 5: Simulated relative yield of rainfed crops under static arrays (GCR 0.1–0.5) against season-mean shading of the crop, baseline climate, held-out sites. The dotted line is proportional loss.

method rise (phenology rule 1.28 against 1.23), and with a response fitted to the field trials the LER of the phenology rule is 1.62. The phenology rule is the best method in all three cases, and light sharing adds LER +0.17, +0.19 and +0.30 over static arrays (conservative, calibrated, field-fitted); only the order of the seasonal and stress rules changes. The absolute gains for rice, wheat and potato are thus conservative with respect to these trials; for maize and soybean the meta-analysis points the other way (Section 3.7).

3.7. A shade-response model learned from field data

Cross-validation by study shows what the data can and cannot support (Tables 8 and 9). Over all 339 shaded observations the root-mean-square error of a held-out study falls from 0.43 for proportional loss $y = 1 - s$ to 0.37 for one curve for all crops and to 0.31 for crop-type curves, but this gain comes from crop types that the simulator does not represent (berries, forages, fruits and vegetables: 0.53 to 0.36). For the four crop types that the simulator uses, the learned curves are not better than proportional loss: the error is 0.222 against 0.220 pooled (difference +0.002, 95% interval -0.023 to $+0.030$), 0.187 against 0.214 for C3 cereals, and slightly worse for maize, grain legumes and tubers. Partial pooling is as good as separate curves (0.312 against 0.313) and the covariates add almost nothing (0.311). Gradient-boosted trees on the same inputs are worse (0.321) than the parametric model. The error of 0.22

for the four crop types is large: yields of a new study differ from the learned curve by about 20 percentage points, which is the between-study spread that a design under uncertainty has to cover. Crop type is the information that transfers between studies, and none of the covariate effects on $\log \beta$ is clearly different from zero (irradiation -0.10 , 90% interval -0.32 to 0.09 ; subtropical zone -0.37 , -0.73 to 0.07 ; intercropping 0.16 , -0.48 to 0.45). The estimated effect of solar-panel shade is not interpretable, because all 46 panel observations are of leafy vegetables, mainly lettuce, and the effect is confounded with crop type.

The learned curves show a clear ordering of crop types (Table 10). At 20% shading C3 cereals and tubers lose less than proportionally (relative yield 0.93), whereas maize (0.73) and grain legumes (0.71) lose more; at 40% shading the values are 0.72 and 0.71 against 0.42 and 0.40. Intervals are wide for maize (0.27–0.59 at 40%) and for tubers, which rest on two studies. The simulator’s responses are close to the data for cereals and potato but, under both calibrated ensembles, too mild for maize at 40% shading (0.75 for E2 against 0.42 in the data); for soybean the E2 value (0.48) lies within the interval of the data (0.40, 80% interval 0.34–0.53). The calibration cannot lower the maize response further, because the simulated shading also relieves heat and drought stress and the compensation and harvest-index parameters are bounded below at zero. The field trials of maize are also milder than the meta-analysis: Ramos-Fuentes et al. (2023) report a harvest index unaffected by shade, and Sekiyama and Nagashima (2019) a yield 5.6% above the control under a low-density stilt-mounted array in one season at one site. For maize and soybean the direction of the bias is uncertain, and the gains reported in this paper are possibly overestimated.

3.8. Learning under shade-response uncertainty

Calibration. Gradient descent through the simulator fitted the five crops to the data-driven curves and the field trials (ensemble E2, Table 11). The data identify two features: the harvest index of maize is hardly affected by shade (multiplier 0.16 ± 0.13 , $c \approx 0.14$), as observed by Ramos-Fuentes et al. (2023), and soybean is more shade sensitive than the cereals (compensation 0.05, harvest-index multiplier 1.66), as in Laub et al. (2022). Wheat has the largest compensation (0.91 ± 0.31) and the largest uncertainty; the compensation is an effective parameter that absorbs diffuse-light and light-saturation effects that the simulator does not model explicitly, not a physical value.

Learned controllers. Controllers were evaluated under four fixed responses on 21 crop–site pairs of split 0 with one training seed, so the intervals below, which are cluster bootstraps over pairs of per-pair mean LER differences, do not include seed or split variation (Table 12 gives season means, which differ slightly). Under every response SALS, trained on the proportional response, is better than the written phenology rule: by +0.019 (+0.007 to +0.039), +0.024 (+0.009 to +0.046), +0.038 (+0.018 to +0.062) and +0.015 (+0.001 to +0.037) LER under the proportional, calibrated, field-fitted and harsh responses. It is also above the grid-tuned window and the CMA-ES optimised window, but by less: 0.005 to 0.018 and 0.005 to 0.010 LER, with intervals that exclude zero for the grid-tuned window under 3 of the four responses and for the CMA-ES window under none. Training on the narrow ensemble E1 did not change this: SALS-robust differs from SALS by -0.005 to $+0.003$ (intervals contain zero except for one response) and SALS-adaptive, which sees the measured biomass ratio, by -0.010 to $+0.007$. Training on the wider ensembles did not help either (the robust controllers fall below SALS even under responses inside their training ensemble, which may point to under-training with our budget of 300 steps): SALS-robust on E2 and E3 and SALS-adaptive on E3 are below SALS by 0.014, 0.017, 0.017 LER under the proportional response, 0.017, 0.021, 0.020 under the calibrated response, 0.004, 0.005, 0.011 under the field-fitted response (intervals contain zero) and 0.008, 0.014, 0.017 under the harsh response; the intervals exclude zero in 7 of the 9 comparisons for the proportional, calibrated and harsh responses. The E3 robust controller is not distinguishable from the written rule ($+0.001$ to $+0.032$) under any response (all intervals contain zero). With one seed and 21 pairs these controllers therefore did not retain the margin of SALS within our training budget; a controller that must perform across plausible responses is less specialised, but a larger budget or more seeds could change this. The feedback rule is worse than the written rule by 0.026–0.058 LER under every response (intervals exclude zero). A plausible reason is that the biomass ratio does not reveal the harvest-index loss from shading around flowering, but we did not test this explanation.

Design under uncertainty. The larger risk is in the design step (Tables 13 and 14, Fig. 6). The designs below are evaluated by leaving one ensemble member out as the truth and using the other four, so that the truth never enters the design. Pooled over the three spatial splits (76 distinct crop–site pairs), a density chosen under the mean response of E2 leaves 70% of the pairs at the floor when the true response is another plausible one,

862 against 87% for an oracle design (E3: 71% and 85%). Choosing the density
 863 so that the floor is met in at least 75% of the other members raises compli-
 864 ance by +7.9 points under E2 (95% cluster-bootstrap interval 4.5 to 11.6)
 865 and +7.9 under E3 (3.9 to 12.1), at a cost of 0.031 and 0.025 in LER. Raising
 866 the floor used in the design from 0.90 to 0.92 under the mean response does
 867 more: +18.0 points (13.4 to 22.6) at 0.068 LER under E2 and +14.3 points
 868 (9.5 to 19.2) at 0.056 under E3, and the shortfall is smaller (mean over pairs
 869 of the maximum of zero and 0.9 minus retention: 0.0022 against 0.0045 for
 870 the quantile rule under E2). The two kinds of rule lie on one frontier (Fig. 6).
 871 Interpolating the margin frontier (floors 0.90 to 0.96) at the compliance of
 872 the ensemble rules, the LER of the margin minus that of the rule is +0.001
 873 (-0.009 to +0.012) for the rule with 75% of the other members and +0.003
 874 (-0.016 to +0.021) for the rule with all of them under E2, and -0.006 (-0.019
 875 to +0.005) and +0.024 (-0.011 to +0.058) under E3 (95% intervals). We find
 876 no detectable difference, but the intervals (down to -0.02 and up to +0.06
 877 LER) are too wide to establish equivalence. The ensemble is thus useful for
 878 sizing the margin (0.02 to 0.04 above the floor for 85–97% of the pairs), not
 879 for replacing it. None of these rules is safe outside the ensemble (Table 14):
 880 under the proportional, calibrated and harsh responses the mean-response
 881 design leaves 7–17% of the pairs at the floor, a floor of 0.92 or the ensemble
 882 rules 12–33%, a floor of 0.94 29–47% and a floor of 0.96 51–78%, at a cost
 883 of about 0.2 LER, whereas the field-fitted response, which lies within the
 884 range spanned by the ensembles, is handled well by every rule (80–99%) ex-
 885 cept the mean-response design (68–79%). Outside the ensemble the shortfall
 886 of the pairs below the floor is also large: under E2 the mean shortfall per
 887 pair (maximum of zero and 0.9 minus retention) is 0.075, 0.060 and 0.104
 888 for the mean-response design under the proportional, calibrated and harsh
 889 responses, against 0.053, 0.038 and 0.078 for a floor of 0.92 and 0.012, 0.007
 890 and 0.024 for a floor of 0.96. Across the five members of an ensemble, the
 891 standard deviation of the simulated relative yield of a crop (static arrays,
 892 baseline climate, held-out sites of split 0) is 0.02 to 0.04 at 20% shading and
 893 0.02 to 0.06 at 40% under E2, and 0.02 to 0.09 and 0.03 to 0.15 under E3
 894 (largest for soybean and potato; ranges between the extreme members up to
 895 0.36), against a leave-one-study-out error of 0.22 for new studies. The two
 896 quantities differ (a spread of curves against an error over observations), but
 897 the ensembles are narrower than the between-study spread for all crops ex-
 898 cept, possibly, soybean under E3. The guarantee therefore holds only within
 899 the range of responses that the ensemble spans; we did not test an ensem-

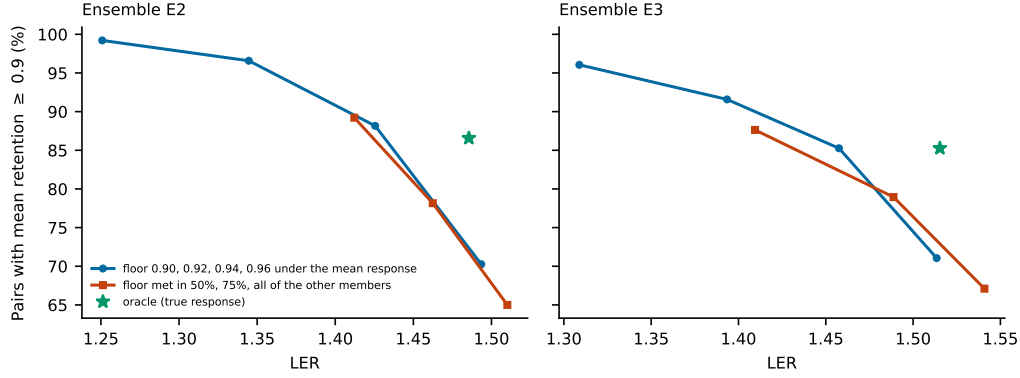


Figure 6: Share of crop–site pairs at the floor against LER for the phenology rule under design rules for shade-response uncertainty (leave-one-member-out, three splits, 76 pairs): a higher floor under the mean response (blue; the first point is the design under the mean response) and a floor met in a share of the other members (orange), with the oracle design (green).

900 ble that spans the proportional and harsh responses, which would protect
 901 against them at a higher cost in LER.

902 3.9. Economics of light sharing

903 All figures in this section follow from the assumed parameters of Table 15
 904 and should be read as a sensitivity analysis, not a forecast. The economics
 905 are dominated by the capacity installed, not by the crop (Table 16): the
 906 annuity and operating cost of the modules is 11–32 kUSD per hectare and
 907 year, whereas the crop revenue lost is about 0.3 kUSD, so a method is worth
 908 what it earns per installed watt. With the densities designed for the floor,
 909 static arrays earn $+0.3 \text{ kUSD ha}^{-1} \text{ yr}^{-1}$ at 60 USD MWh^{-1} and break even
 910 at 59, whereas all light-sharing methods lose money at that price (-3 to
 911 -10 kUSD) and break even at 70 (stress rule), 73 (SALS), 73 (CMA-ES
 912 window), 77 (grid-tuned window), 80 (written rule) and 92 (seasonal sharing)
 913 USD MWh^{-1} . The 95% cluster-bootstrap intervals over pairs are 56–61 for
 914 static arrays, 69–76 for SALS, 70–77 for the CMA-ES window and 75–84 for
 915 the written rule, and 46–50 for the plant (break-even prices of SALS and the
 916 CMA-ES window differ by -0.4 , -1.1 to 0.4); they cover the sampling of
 917 pairs and not the cost assumptions. At 90 USD MWh^{-1} static arrays (6.3
 918 kUSD) are better than the written (4.0) and grid-tuned (5.5) rules, close
 919 to the CMA-ES window (7.3) and worse than SALS (7.6), which overtakes

static arrays above about 85 USD MWh⁻¹; the difference between SALS and static arrays at 90 USD MWh⁻¹ is +1.3 kUSD (95% interval -0.04 to 2.7, 76 pairs) and not distinguishable from zero, whereas SALS exceeds the CMA-ES window by 0.3 (-0.1 to 0.8, not distinguishable from zero), the grid-tuned window by 2.1 (1.6 to 2.7) and the written rule by 3.6 (2.8 to 4.5). A conventional plant that replaces the crop breaks even at 48 USD MWh⁻¹ and is more profitable than every agrivoltaic option above that price (16 kUSD at 60, 94 kUSD at 120, against -6 and 21 for SALS); below it all options lose money and static arrays lose least (at 30 USD MWh⁻¹: plant -23.2, SALS -18.7, static -5.7 kUSD). Per megawatt-hour, the electricity of SALS costs about 73 USD including the crop lost, against 48 USD for the plant, a difference of 25 USD MWh⁻¹. A premium of that size would equalise the unit cost but not the profit per hectare, because a hectare of plant produces three times as much electricity (1306 against 438 MWh): for the same electricity as one hectare of plant, SALS needs about three hectares of array, which costs about 25 USD MWh⁻¹ more including the crop lost but keeps about 90% of the yield of that cropland; relative to the plant, SALS would need a premium of about 49 USD MWh⁻¹ at a market price of 60 and about 109 at 90 to match its profit per hectare, and only about 13 at 60 (none above 73) to match the open field. The sensitivity is large (Table 17), and the ranking of static arrays and light sharing flips with the module cost (at 1.5 USD Wp⁻¹ static arrays are better than SALS, at 0.7 SALS leads by 7 kUSD): at 90 USD MWh⁻¹ the net value of SALS changes from +17 to -8 kUSD between module costs of 0.7 and 1.5 USD Wp⁻¹, whereas crop prices matter little.

Because the density was designed for the floor and not for profit, we also let each site choose, among the densities that meet the floor in its history, the one that maximises its historical net value (sites where none pays stay open field; Fig. 7). Then no method pays at 30–40 USD MWh⁻¹; at 60 14–59% of the pairs install an array (none for seasonal sharing) and the net values are small (static arrays 0.9, SALS 0.3 kUSD); static arrays are the best option up to about 75 USD MWh⁻¹, SALS overtakes them at about 75 (4.9 against 4.3 at 80, 8.3 against 6.3 at 90, 20.7 against 12.3 at 120), the CMA-ES window at about 78, the stress rule at about 81, the grid-tuned window at about 87 and the written rule at about 97. The value of control is therefore real but depends on a high price: at 90 USD MWh⁻¹ and with profit-optimal densities SALS exceeds the CMA-ES window by 0.4 kUSD ha⁻¹ yr⁻¹ (95% cluster-bootstrap interval 0.01 to 0.8), the grid-tuned window by 1.7 (1.2 to 2.2), the written rule by 2.8 (2.0 to 3.6) and static arrays by 2.0 (0.8 to 3.3);

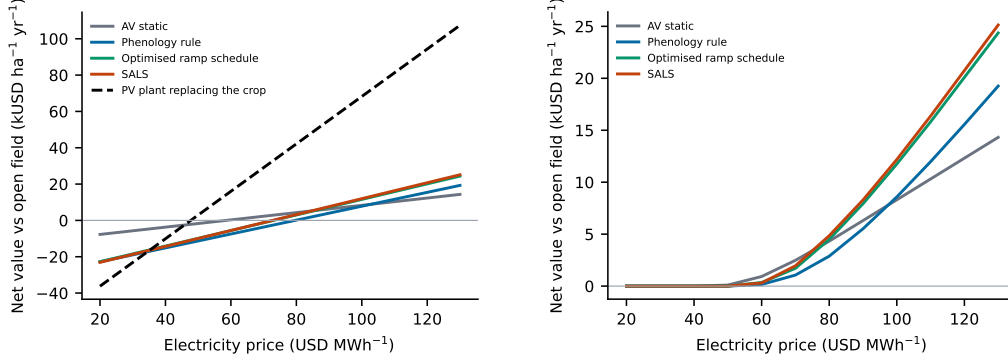


Figure 7: Net value per hectare relative to the open field against electricity price. Left: densities designed for the floor, with a conventional plant replacing the crop (dashed). Right: each site chooses the feasible density that maximises its historical net value. Illustrative parameters of Table 15.

the CMA-ES window exceeds static arrays by about 1.6 (0.4 to 2.8), so most of the value of control is obtained with a well-optimised rule.

Cost of the safety margin. At 90 USD MWh⁻¹, moving from a density designed under the mean response to a floor of 0.92 under that response costs 1.00 kUSD ha⁻¹ yr⁻¹ (17%) under E2 for 18 more points of compliance (70% to 88%) and 1.13 kUSD (20%) under E3 for 14 points (71% to 85%), about 56 and 79 USD ha⁻¹ yr⁻¹ per point. The ensemble rule that requires all other members to meet the floor costs 0.85 and 1.22 kUSD for 19 and 17 points (45 and 74 USD per point), so on this measure it is cheaper than the margin under E2 and similar under E3. At 60 USD MWh⁻¹ a margin has a negative cost, because every module beyond break-even loses money; the economics of the design margin therefore depend on the price level, whereas its effect on compliance does not.

3.10. Agrivoltaics under warming

Under warming, the open-field production of the held-out sites (Fig. 8) falls at +3 °C by 23% for maize and 18% for rice, is nearly unchanged for soybean, and rises for wheat (+3%) and potato (+10%). Agrivoltaic production with light-sharing control follows the open field at about 90% of its level, so it inherits the same climate impacts (maize -31%, rice -27% relative to the baseline open field at +3 °C). Agrivoltaics does not protect staple production against warming in this model. The LER of light-sharing systems increases

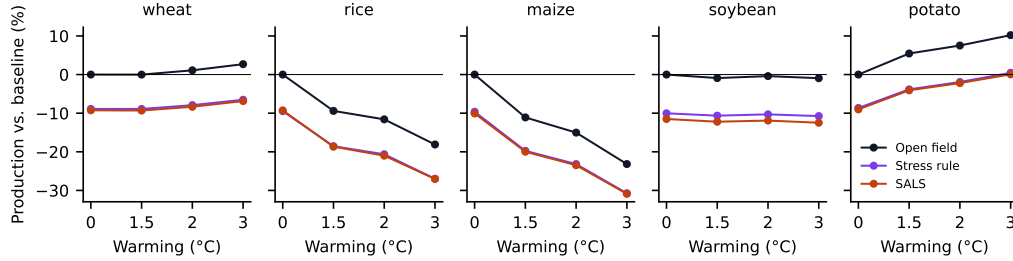


Figure 8: Production relative to the baseline open field (sum over held-out site-seasons) as a function of warming, for the open field, the stress rule and SALS.

slightly with warming (SALS 1.270 to 1.297; phenology rule 1.236 to 1.265), because the floor allows slightly denser arrays in warmer conditions (GCR 0.173 to 0.177 and electricity 510 to 540 MWh ha⁻¹ yr⁻¹). The margin of SALS over the rules does not grow with warming (Table A.18).

3.11. Global potential and what it would take

Every site is evaluated when it is held out (222 of the 400 crop–site pairs are held out in at least one split, and three of them have no valid season, leaving 219), and each crop’s mean electricity per hectare is scaled to its physical area in SPAM 2010 (581 Mha for the five crops, which avoids counting multiple-cropped land twice). Equipping 5% of this area (29.1 Mha) with SALS arrays at the floor would produce 14.4 PWh yr⁻¹ under the baseline climate and 16.0 PWh yr⁻¹ at +3 °C, with a production-weighted retention of 0.903 and 0.900. The phenology rule would give 13.0 and 14.5 PWh yr⁻¹, and static arrays 6.4 and 6.9 PWh yr⁻¹. For comparison, world electricity generation was about 30 PWh in 2023 (Ember, 2024).

These numbers require large amounts of hardware. SALS at 5% coverage needs 10.6 TWp of modules (11.3 TWp at +3 °C), at a capacity factor of 15.5% (16.2%). World solar PV capacity at the end of 2024 was about 1.9 TW (IRENA, 2025), so the requirement is about 5.6 times the installed PV fleet. The electricity is also variable, peaks at midday and would be curtailed or stored. The figures are technical potentials, not projections. Ordering sites by electricity per hectare gives a supply curve (Fig. 9): the best 2% of the staple cropland would supply about 21 PWh yr⁻¹ and the best 4.7% about 40 PWh yr⁻¹ with SALS. The best tenth of the sites are mostly rainfed maize and soybean sites in the Americas and in southern and eastern Africa (7% mean irrigated fraction against 29% overall), which are also

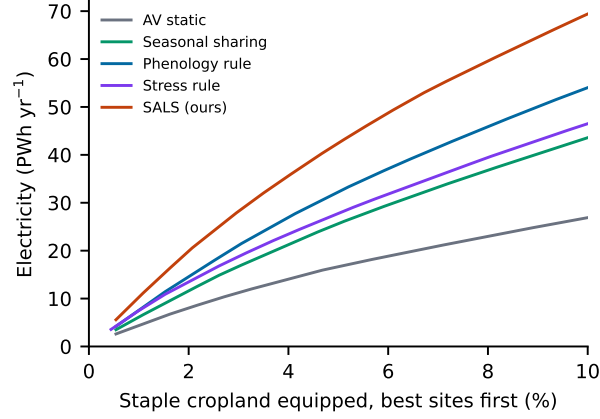


Figure 9: Supply curve under the baseline climate: cumulative electricity when sites are equipped in order of decreasing electricity per hectare, against the share of staple cropland (physical area) equipped. Each controller is evaluated at the density that keeps its mean retention at 0.9.

among the regions where the floor is hardest to meet (Section 3.5). Siting by electricity alone would therefore concentrate agrivoltaics where food-security risk is highest.

Maize and soybean, for which the simulated shade response is milder than the meta-analysis (Section 3.7), account for 53% of the electricity of the upscaling (a scaling convention of 5% of the cropland, reported without an interval) (maize 30%, soybean 23%, wheat 29%, rice 15%, potato 3%). Counting only wheat, rice and potato would give about 7 PWh yr⁻¹ at the same coverage, so the global figure should be read as an upper estimate for about half of its content. By crop, the LER of SALS under the baseline climate is 1.20 for wheat and rice, 1.31 for potato, 1.30 for maize and 1.41 for soybean.

4. Discussion

4.1. What the results establish

Four findings are discussed; the first three stand on consistent evidence and the fourth is not established. First, sharing light with the crop is the main lever on the electricity that tracker-based agrivoltaic arrays can produce under a yield floor. With static trackers the floor limits arrays to about 6% ground coverage and an LER of 1.07. Any light-sharing control allows 2.1

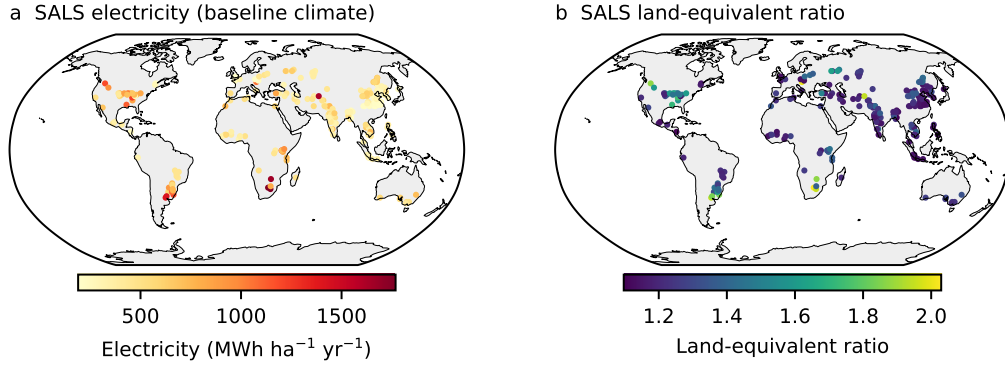


Figure 10: SALS electricity (a) and land-equivalent ratio (b) at the pooled held-out sites (baseline climate, mean of seasons 2015–2019).

to 2.9 times the density, and the LER rises to 1.18–1.24 for hand-written rules. Second, the best simple rule follows crop phenology. It shares light from the onset of canopy development (5% of the thermal-time requirement) until maturity and produces electricity before and after, when the crop does not use the light. It needs only the crop’s thermal time, which is observable on the farm, and it is insensitive to forecast errors, whereas the rule that reacts to heat and drought stress is worse in every climate and degrades with forecast error. Third, the qualitative result does not depend on the weakest part of the model. The simulated shade response is conservative against the field trials we could use (Section 3.6); calibrating it raises the allowed density, and the phenology rule stays the best method, with a gain over static arrays of 0.17, 0.19 and 0.30 LER under the conservative, calibrated and field-fitted responses. However, the response learned from the meta-analysis shows steeper losses for maize and grain legumes than the simulator reproduces (Section 3.7), so the absolute gains are conservative for wheat, rice and potato and uncertain, possibly overestimated, for maize and soybean (53% of the upscaled electricity). Fourth, a controller learned through the differentiable simulator is better than the best rule we could write, by 0.044 in LER against the written rule but by only 0.007 (0.002 to 0.013; 2.3% more electricity at equal retention, and not distinguishable from zero at matched retention) against a window optimised on the training seasons of each split. The margin over the written rule is consistent over training seeds (main split, range 0.002 in LER), the three spatial splits, three canopy-cooling

assumptions and the baseline and +3 °C climates, whereas the margin over the optimised window is small and varies between splits (0.007, 0.010 and 0.005 LER on splits 0, 1 and 2; per-pair medians 0.001 to 0.002; one training seed per split), so it is not established as a reproducible gain. This gain is small compared with the gap between static and dynamic control.

4.2. What role does artificial intelligence play?

We tested three ways in which learning could matter, and the answers differ. The first is learning the crop’s response to shade from data. The learned model is useful mainly as a measurement of how little can be learned: a hierarchical model fitted to 428 observations from 58 studies predicts a held-out study of the four staple crop types with an error of 0.22 in relative yield, no better than proportional loss, and the gain over proportional loss (0.43 to 0.31 over all types) comes from crop types that the simulator does not represent. Crop type is the information that transfers, whereas irradiation, climate zone and shade type add nothing measurable and a non-parametric model (gradient-boosted trees) is worse with this number of studies (Section 3.7). Two things follow. The shade response of the staple crops is a distribution with a spread of about 20 percentage points between studies and not a curve, and the process-based simulator does not reproduce the steep losses of maize and legumes seen in the data. Learning the response was limited by the data and not by the model: two studies support the tuber curve, no rice study exists, and 46 panel observations are all leafy vegetables.

The second is calibration through the differentiable simulator. Gradient descent through whole seasons fitted the response of each crop to the learned curves and to the field trials, and it produced an ensemble of plausible responses that fits the trials and the curves for cereals and potato but not the maize and soybean curves (Section 3.8). The ensemble informs the design decision: a density chosen under the mean response leaves about 70% of crop–site pairs at the floor if another plausible response holds, against 85–87% for an oracle, and a density chosen to meet the floor in 75% of plausible responses recovers about half of that gap at a cost of 0.03 LER. A simple margin reaches similar compliance: raising the floor from 0.90 to 0.92 under the mean response reaches 85–88% of the pairs, and interpolating the margin frontier at the compliance of the ensemble rules gives LER differences (margin minus rule) of +0.001 (−0.009 to 0.012) and +0.003 (−0.016 to 0.021) under E2 and −0.006 (−0.019 to 0.005) and +0.024 (−0.011 to 0.058) under E3, so the two are not distinguishable on this sample (equivalence is not

1084 established, since the intervals reach ± 0.02 to 0.06 LER). None of the rules
 1085 is safe outside the ensemble (7–33% of the pairs at the floor under the pro-
 1086 portional, calibrated and harsh responses for the mean design, the margin of
 1087 0.92 and the ensemble rules), and only a floor of 0.96 reaches 51–78% at a
 1088 cost of about 0.2 LER. What the ensemble adds is the size of the margin,
 1089 not a better rule. The quantile rule is a scenario-type design (Calafiore and
 1090 Campi, 2006; Campi and Garatti, 2008); with five members it has none of
 1091 the finite-sample guarantees of that theory, and since the comparison with
 1092 the margin shows no detectable difference (with intervals too wide to estab-
 1093 lish equivalence), we see no evidence that it is needed. The bottleneck of
 1094 tracker-based agrivoltaics under a yield-retention floor is the uncertainty of
 1095 the crop response that enters the design, and it is data on that response, not
 1096 larger networks, that would improve the result; the learned curves show a
 1097 spread of about 20 percentage points between studies. The guarantee holds
 1098 only within the range of responses that the ensemble spans.

1099 The third is control, and here learning contributes little. Differentiabil-
 1100 ity is what lets a network of 5441 weights be trained in 300 gradient steps
 1101 through whole seasons; a derivative-free search such as CMA-ES handled a
 1102 five-parameter schedule with 200 simulator evaluations per run and found an
 1103 almost binary window that is only 0.007 LER behind SALS, but would not
 1104 scale to the network. The learned controller adds 0.007 LER over an opti-
 1105 mised window (and 0.017 over a grid-tuned one, 0.044 over the written rule),
 1106 and 0.005 – 0.010 over the optimised window under the four alternative re-
 1107 sponses on one split (21 pairs, intervals containing zero); a reader who needs
 1108 a simple, auditable controller can use that window and obtain about 84% of
 1109 the LER gain of SALS over the written rule. Neither of the two extensions
 1110 that we expected to matter under uncertainty did: training on ensembles of
 1111 responses (E1, E2 and E3) did not help and, for the wider ensembles, did
 1112 not retain the margin over the written rule within our training budget (one
 1113 seed, 21 pairs), and neither did giving the controller a farm-measurable sig-
 1114 nal, the biomass of the crop relative to a reference plot, nor a feedback rule
 1115 built on that signal. The learned policy also has to be initialised from the
 1116 phenology rule (Section 4.5). We therefore do not claim that learned control
 1117 is what makes agrivoltaics work: in this model, the schedule that a learned
 1118 policy converges to is a smooth function of crop phenology that an optimised
 1119 window already approximates to within 0.007 LER, and the source of the
 1120 remaining margin is not identified.

1121 4.3. *What the results do not establish*

1122 The early hypothesis that stress-aware control, that is, shading the crop
1123 on heat and drought days, would be a main source of benefit under warming
1124 was not supported. The stress rule performs worse than the phenology rule
1125 in every climate, and a learned controller without stress inputs performs as
1126 well as one with them. A plausible reason is that in our crop model the yield
1127 gain from shading on a hot or dry day is smaller than the loss of light on
1128 that day, so stress information does not help. Whether this holds for crop
1129 models with more detailed canopy-temperature and heat-stress responses is
1130 an open question. Agrivoltaics does not protect staple production against
1131 warming in our simulations: light-sharing systems produce about 90% of the
1132 open-field yield in every climate and follow the same decline in maize and
1133 rice (Fig. 8).

1134 4.4. *Food security in practice*

1135 Holding the historical mean retention at 90% does not hold it in each
1136 season or at each site. 70% of SALS seasons meet the floor, the lower tail is
1137 slightly heavier than for the rules, and production-weighted retention is below
1138 0.9 for soybean (0.888) and maize (0.898) and in several regions. Chance-
1139 constrained design raises season-level compliance to about 92% at a cost of
1140 0.065 in LER, and the ranking of the controllers is preserved. Whichever for-
1141 mulation is chosen, a per-site regulatory floor would need a region- and crop-
1142 specific design step. The upscaled potential (Section 3.11) shows the trade-off
1143 in magnitude. The 14 PWh yr⁻¹ that 5% of the staple cropland could supply
1144 need about 10 TWp of modules, several times the present global fleet, and
1145 the best sites are rainfed maize and soybean areas where production-weighted
1146 retention is lowest.

1147 **Who bears the risk.** A floor defined on the multi-year mean leaves
1148 the farmer exposed in the seasons below it: with the density designed for
1149 the floor, about 30% of seasons fall short on the main split (70% compliance
1150 for SALS), the shortfall is mostly small (2.7% of SALS seasons are below
1151 0.8 of the open-field yield and the 10th percentile of retention is 0.883), and
1152 a per-season guarantee through chance-constrained design costs 0.065 LER.
1153 Siting by electricity favours maize and soybean areas, where compliance is
1154 lowest (Section 3.11), so the risk concentrates where the energy gain is largest.
1155 The crop revenue at stake (about 0.3 kUSD ha⁻¹ yr⁻¹) is small against 20–30
1156 kUSD of electricity revenue, so the exposure is mainly one of compliance with
1157 the floor and of the contract, and it coincides with the largest uncertainty

of the shade response, that of maize and soybean. Who carries it, through contracts, insurance or a premium, is a design question that the model does not address, and food is valued only through crop revenue at market prices.

4.5. Limitations

- *Simulation only.* All results come from a simulator. PV output was benchmarked against PVGIS (29 sites, mean bias -4.9%), but the crop response to shade was learned from one meta-analysis data set and compared with four field trials (Sections 3.6 and 3.7). The simulated response is conservative with respect to the field trials of rice, wheat and potato; for maize it is conservative with respect to the trials but milder than the meta-analysis, and for soybean the simulator (0.48 at 40% shading) lies within the interval of the data (0.40, 80% interval 0.34–0.53), while for maize the data give 0.42 against 0.75 in the simulator, so for these crops the direction of the bias depends on the source. The harvest-index sensitivities h are assumptions, and the sensitivity analyses use subsamples of crop–site pairs (22 for the rule comparison under alternative responses, 21 for the controller comparison, one training seed).
- *Canopy cooling.* The effect of shade on canopy temperature was represented by one coefficient κ , applied to a heat-stress function that was calibrated on air temperature (Siebert et al., 2014; Webber et al., 2018). The ranking of the controllers is robust to κ between 0 and 2°C , but the absolute electricity changes by about $\pm 8\%$. Nocturnal warming under modules and rain redistribution are not modelled.
- *Crop model.* SIMPLE has a fixed harvest index and linear radiation-use efficiency, and its CO_2 response is that of controlled experiments that tend to overstate field responses (Long et al., 2006). More detailed models with grain-number formation and canopy energy balance may favour different schedules.
- *Layouts and models.* The baseline of static arrays is a backtracking tracker held at the energy-optimal angle; flat and 30° -rotated (east-facing, not equator-facing) fixed arrays did no better in a check on 21 pairs, but vertical bifacial layouts (Willockx et al., 2023; Campana et al., 2021) are not modelled, and the ground coverage ratio is a coarse proxy of crop productivity (Dupraz, 2023). The PV model is isotropic

1193 in the diffuse component and has no bifacial gain; it was benchmarked
1194 against PVGIS at low density only and has not been compared with
1195 a reference implementation such as pvlb (F. Holmgren et al., 2018).
1196 The rotation limit was varied between 45 and 80 degrees and changes
1197 the size of the benefit but not the ranking; the harvest-index window
1198 is assumed to be common to all crops, and shifting it or removing the
1199 penalty did not change the ranking of the rules.

- 1200 • *Climate scenarios.* Warming was applied as uniform temperature shifts
1201 with fixed precipitation, radiation and relative humidity, and with un-
1202 changed cultivars and sowing dates. It is not a bias-adjusted climate
1203 projection.
- 1204 • *Initialisation and state information.* The network is initialised from the
1205 phenology rule, because training from random weights collapsed to full
1206 sharing. The gain of SALS is therefore the gain of fine-tuning that rule,
1207 and we cannot say whether a better optimiser or a richer policy class
1208 would find larger gains. The controller uses the simulated state of the
1209 crop and soil (thermal time, canopy interception, soil water) without
1210 measurement error, and the forecast errors we tested apply to weather
1211 only.
- 1212 • *Economics and deployment.* The economic analysis (Section 3.9) uses
1213 assumed prices and costs, no grid, curtailment, storage or subsidy
1214 model, and does not value food security; its break-even prices move
1215 with the capex and the discount rate (Fig. 7). The value of midday
1216 electricity, curtailment, storage, structural cost and farmer acceptance
1217 (Trommsdorff et al., 2021) will determine the realisable share. The av-
1218 erage row pitch at the chosen densities (about 13 m for 2.2-m modules)
1219 leaves room for machinery, but the densest designs (GCR up to 0.5)
1220 have a pitch of only 4.4 m.

1221 5. Conclusions

1222 Light sharing, the daily choice between maximum electricity and letting
1223 light through to the crop, is the main lever on the electricity of tracker-based
1224 agrivoltaics at a given yield-retention floor. In a differentiable simulator,
1225 evaluated on held-out crop–site pairs (92 on the main split, 76 distinct pairs
1226 pooled over three splits for the controller and design comparisons) with the

density of each site chosen from its own history, light-sharing rules allowed 2.1 to 2.9 times the module density of static arrays at a mean yield retention of 90% and raised the land-equivalent ratio from 1.07 to 1.18–1.24 under the proportional-growth shade response. A rule that follows crop phenology and needs no weather forecast was the best written rule. The lever pays only at high electricity prices: with illustrative costs it breaks even with the open field, at floor-designed densities, at 70 to 92 USD MWh⁻¹ (73 for the learned controller, 59 for static arrays), static arrays earn more below about 75 to 85 USD MWh⁻¹, and a conventional plant replacing the crop is more profitable above its break-even of 48 USD MWh⁻¹. In this model agrivoltaics therefore pays only with a premium, land-use rules or a value for food production that the model does not include.

The crop’s response to shade, the least certain input, was learned from 428 field observations. For the staple crop types a hierarchical model predicted a held-out study no better than proportional loss (error 0.22), which measures the between-study spread, and gradient descent through the simulator turned the learned curves and four field trials into ensembles of plausible responses. The simulator is conservative against the trials for rice, wheat and potato; for maize and soybean, which supply 53% of the upscaled electricity, the meta-analysis indicates larger losses than simulated and one maize trial smaller ones, so the gains for these crops are uncertain and possibly overestimated. A learned controller exceeded the written phenology rule by 0.044 LER but a CMA-ES-optimised window by only 0.007 (95% interval 0.002–0.013, one training seed per split; at matched retention 0.004, interval –0.0003 to 0.007). The source of this margin is not identified: it is not explained by the tuning of a window or a ramp, and masking soil water, aridity and forecasts leaves the controller unchanged. Training on ensembles of shade responses, or giving the controller a farm-measurable biomass signal, did not help within our budget (one seed, 21 pairs).

Compliance depends more on the response assumed in the design than on the controller. A density chosen under the mean response left about 70% of crop–site pairs at the floor when another plausible response held, against 85–87% for an oracle design; this gap of 15 points (a model-misspecification effect) is larger than the 4 points between light-sharing controllers at a fixed design rule under a known response, but the two are not like-for-like. Designing for the floor in 75% of the plausible responses recovered about half of the gap at a cost of 0.03 LER, and a floor raised by 0.02 under the mean response gave no detectable difference at equal compliance (the intervals in-

1265 clude zero and are too wide to establish equivalence); no rule was safe outside
1266 the ensemble, whose members differ less from one another than a new study
1267 differs from the learned curve. The ensemble is therefore useful to size a
1268 safety margin, not to replace it. The yield floor is met on average but not in
1269 every season or region, and the electricity potential of 5% of staple cropland
1270 (about 14 PWh yr⁻¹, a scaling convention without an interval) requires about
1271 10 TWp of modules. Future work should validate the crop response to shade
1272 under trackers with more field data, notably for maize and legumes where the
1273 meta-analysis and the trials disagree, extend the crop model to grain-number
1274 formation and canopy energy balance, add costs and grid constraints, and
1275 pursue controllers that guarantee the yield-retention floor per site and per
1276 season.

1277 **CRedit authorship contribution statement**

1278 **Quang-Vinh Dang:** Conceptualization, Methodology, Software, Formal
1279 analysis, Investigation, Data curation, Visualization, Writing – original draft,
1280 Writing – review & editing.

1281 **Declaration of competing interest**

1282 The author has no known competing financial interests or personal re-
1283 lationships that could have appeared to influence the work reported in this
1284 paper.

1285 **Data and code availability**

1286 All data used are public: SPAM 2010 v2r0 harvested areas (Yu et al.,
1287 2020), the GGCM Phase 3 crop calendar (Jägermeyr et al., 2021), NASA
1288 POWER daily weather (Sparks, 2018) the SIMPLE crop parameters (Zhao
1289 et al., 2019) and the shade-response meta-analysis data of Laub et al. (2022)
1290 (Zenodo 10.5281/zenodo.5716091, CC BY 4.0, copied into the repository).
1291 The code of the simulator, of SALS and of all experiments, together with the
1292 site list, is available from the author’s public repository (<https://github.com/vinhqddang/Artificial-Intelligence-in-Agriculture>); a versioned
1293 archive with a DOI will be deposited at acceptance.
1294

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Appendix A. Statistical comparison

References

- Adeh, E.H., Good, S.P., Calaf, M., Higgins, C.W., 2019. Solar pv power potential is greatest over croplands. *Scientific Reports* 9, 11442. doi:10.1038/s41598-019-47803-3.
- Allen, R.G., Pereira, L.S., Raes, D., Smith, M., 1998. Crop evapotranspiration: Guidelines for computing crop water requirements. *FAO Irrigation and Drainage Paper* 56. Food and Agriculture Organization of the United Nations. Rome.
- Amaducci, S., Yin, X., Colauzzi, M., 2018. Agrivoltaic systems to optimise land use for electric energy production. *Applied Energy* 220, 545–561. doi:10.1016/j.apenergy.2018.03.081.
- Andrade, F.H., Vega, C., Uhart, S., Cirilo, A., Cantarero, M., Valentinuz, O., 1999. Kernel number determination in maize. *Crop Science* 39, 453–459. doi:10.2135/cropsci1999.0011183X0039000200026x.
- Barron-Gafford, G.A., Pavao-Zuckerman, M.A., Minor, R.L., Sutter, L.F., Barnett-Moreno, I., Blackett, D.T., Thompson, M., Dimond, K., Gerlak, A.K., Nabhan, G.P., Macknick, J.E., 2019. Agrivoltaics provide mutual benefits across the food–energy–water nexus in drylands. *Nature Sustainability* 2, 848–855. doi:10.1038/s41893-019-0364-5.
- Calafiore, G.C., Campi, M.C., 2006. The scenario approach to robust control design. *IEEE Transactions on Automatic Control* 51, 742–753. doi:10.1109/TAC.2006.875041.
- Campana, P.E., Stridh, B., Amaducci, S., Colauzzi, M., 2021. Optimisation of vertically mounted agrivoltaic systems. *Journal of Cleaner Production* 325, 129091. doi:10.1016/j.jclepro.2021.129091.

1324 Campi, M.C., Garatti, S., 2008. The exact feasibility of randomized solutions
 1325 of uncertain convex programs. *SIAM Journal on Optimization* 19, 1211–
 1326 1230. doi:10.1137/07069821X.

1327 Collares-Pereira, M., Rabl, A., 1979. The average distribution of solar
 1328 radiation-correlations between diffuse and hemispherical and between daily
 1329 and hourly insolation values. *Solar Energy* 22, 155–164. doi:10.1016/
 1330 0038-092x(79)90100-2.

1331 DIN, 2021. DIN SPEC 91434:2021-05: Agri-photovoltaic systems – require-
 1332 ments for primary agricultural use. Beuth Verlag, Berlin.

1333 Dinesh, H., Pearce, J.M., 2016. The potential of agrivoltaic systems. *Renew-
 1334 able and Sustainable Energy Reviews* 54, 299–308. doi:10.1016/j.rser.
 1335 2015.10.024.

1336 Dupraz, C., 2023. Assessment of the ground coverage ratio of agrivoltaic
 1337 systems as a proxy for potential crop productivity. *Agroforestry Systems*
 1338 98, 2679–2696. doi:10.1007/s10457-023-00906-3.

1339 Dupraz, C., Marrou, H., Talbot, G., Dufour, L., Nogier, A., Ferard, Y., 2011.
 1340 Combining solar photovoltaic panels and food crops for optimising land
 1341 use: Towards new agrivoltaic schemes. *Renewable Energy* 36, 2725–2732.
 1342 doi:10.1016/j.renene.2011.03.005.

1343 Elamri, Y., Cheviron, B., Lopez, J.M., Dejean, C., Belaud, G., 2018. Water
 1344 budget and crop modelling for agrivoltaic systems: Application to irrigated
 1345 lettuces. *Agricultural Water Management* 208, 440–453. doi:10.1016/j.
 1346 agwat.2018.07.001.

1347 Ember, 2024. Global Electricity Review 2024. Technical Report. Ember.
 1348 London.

1349 European Commission, Joint Research Centre, 2022. Photovoltaic geo-
 1350 graphical information system (PVGIS), version 5.2. [https://re.jrc.ec.
 1351 europa.eu/pvg_tools/](https://re.jrc.ec.europa.eu/pvg_tools/).

1352 F. Holmgren, W., W. Hansen, C., A. Mikofski, M., 2018. pvlib python: a
 1353 python package for modeling solar energy systems. *Journal of Open Source
 1354 Software* 3, 884. doi:10.21105/joss.00884.

1355 Fischer, R.A., 1985. Number of kernels in wheat crops and the influence of
1356 solar radiation and temperature. *The Journal of Agricultural Science* 105,
1357 447–461. doi:10.1017/S0021859600056495.

1358 Gautron, R., Maillard, O.A., Preux, P., Corbeels, M., Sabbadin, R., 2022.
1359 Reinforcement learning for crop management support: Review, prospects
1360 and challenges. *Computers and Electronics in Agriculture* 200, 107182.
1361 doi:10.1016/j.compag.2022.107182.

1362 Gonocruz, R.A., Nakamura, R., Yoshino, K., Homma, M., Doi, T., Yoshida,
1363 Y., Tani, A., 2021. Analysis of the rice yield under an agrivoltaic
1364 system: A case study in japan. *Environments* 8, 65. doi:10.3390/
1365 environments8070065.

1366 Hansen, N., Ostermeier, A., 2001. Completely derandomized self-adaptation
1367 in evolution strategies. *Evolutionary Computation* 9, 159–195. doi:10.
1368 1162/106365601750190398.

1369 Hassanpour Adeg, E., Selker, J.S., Higgins, C.W., 2018. Remarkable agri-
1370 voltaic influence on soil moisture, micrometeorology and water-use effi-
1371 ciency. *PLOS ONE* 13, e0203256. doi:10.1371/journal.pone.0203256.

1372 IPCC, 2021. *Climate Change 2021: The Physical Science Basis. Contribution*
1373 *of Working Group I to the Sixth Assessment Report of the Intergovern-*
1374 *mental Panel on Climate Change.* Cambridge University Press, Cambridge.
1375 doi:10.1017/9781009157896.

1376 IRENA, 2025. *Renewable Capacity Statistics 2025.* Technical Report. Inter-
1377 national Renewable Energy Agency. Abu Dhabi.

1378 Jägermeyr, J., Müller, C., Ruane, A.C., Elliott, J., Balkovic, J., Castillo, O.,
1379 Faye, B., Foster, I., Folberth, C., Franke, J.A., Fuchs, K., Guarin, J.R.,
1380 Heinke, J., Hoogenboom, G., Iizumi, T., Jain, A.K., Kelly, D., Khabarov,
1381 N., Lange, S., Lin, T.S., Liu, W., Mialyk, O., Minoli, S., Moyer, E.J.,
1382 Okada, M., Phillips, M., Porter, C., Rabin, S.S., Scheer, C., Schneider,
1383 J.M., Schyns, J.F., Skalsky, R., Smerald, A., Stella, T., Stephens, H.,
1384 Webber, H., Zabel, F., Rosenzweig, C., 2021. Climate impacts on global
1385 agriculture emerge earlier in new generation of climate and crop models.
1386 *Nature Food* 2, 873–885. doi:10.1038/s43016-021-00400-y.

- 1387 Karniadakis, G.E., Kevrekidis, I.G., Lu, L., Perdikaris, P., Wang, S., Yang,
1388 L., 2021. Physics-informed machine learning. *Nature Reviews Physics* 3,
1389 422–440. doi:10.1038/s42254-021-00314-5.
- 1390 Kingma, D.P., Ba, J., 2015. Adam: A method for stochastic optimization,
1391 in: *International Conference on Learning Representations*.
- 1392 Laub, M., Pataczek, L., Feuerbacher, A., Zikeli, S., Högy, P., 2022. Contrast-
1393 ing yield responses at varying levels of shade suggest different suitability
1394 of crops for dual land-use systems: a meta-analysis. *Agronomy for Sus-
1395 tainable Development* 42, 51. doi:10.1007/s13593-022-00783-7.
- 1396 Liu, B.Y., Jordan, R.C., 1960. The interrelationship and characteristic dis-
1397 tribution of direct, diffuse and total solar radiation. *Solar Energy* 4, 1–19.
1398 doi:10.1016/0038-092x(60)90062-1.
- 1399 Long, S.P., Ainsworth, E.A., Leakey, A.D.B., Nösberger, J., Ort, D.R., 2006.
1400 Food for thought: Lower-than-expected crop yield stimulation with ris-
1401 ing CO₂ concentrations. *Science* 312, 1918–1921. doi:10.1126/science.
1402 1114722.
- 1403 Lorenzo, E., Narvarte, L., Muñoz, J., 2011. Tracking and back-tracking.
1404 *Progress in Photovoltaics: Research and Applications* 19, 747–753. doi:10.
1405 1002/pip.1085.
- 1406 Mamun, M.A.A., Dargusch, P., Wadley, D., Zulkarnain, N.A., Aziz, A.A.,
1407 2022. A review of research on agrivoltaic systems. *Renewable and Sustain-
1408 able Energy Reviews* 161, 112351. doi:10.1016/j.rser.2022.112351.
- 1409 Marchau, V.A.W.J., Walker, W.E., Bloemen, P.J.T.M., Popper, S.W. (Eds.),
1410 2019. *Decision Making under Deep Uncertainty: From Theory to Practice*.
1411 Springer. doi:10.1007/978-3-030-05252-2.
- 1412 Marrou, H., Dufour, L., Wery, J., 2013a. How does a shelter of solar panels
1413 influence water flows in a soil–crop system? *European Journal of Agron-
1414 omy* 50, 38–51. doi:10.1016/j.eja.2013.05.004.
- 1415 Marrou, H., Guillioni, L., Dufour, L., Dupraz, C., Wery, J., 2013b. Microcli-
1416 mate under agrivoltaic systems: Is crop growth rate affected in the partial
1417 shade of solar panels? *Agricultural and Forest Meteorology* 177, 117–132.
1418 doi:10.1016/j.agrformet.2013.04.012.

- 1419 Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J., Chanan, G., Killeen,
1420 T., Lin, Z., Gimelshein, N., Antiga, L., et al., 2019. PyTorch: An impera-
1421 tive style, high-performance deep learning library, in: Advances in Neural
1422 Information Processing Systems, pp. 8024–8035.
- 1423 Ramos-Fuentes, I.A., Elamri, Y., Cheviron, B., Dejean, C., Belaud, G.,
1424 Fumey, D., 2023. Effects of shade and deficit irrigation on maize growth
1425 and development in fixed and dynamic agrivoltaic systems. *Agricultural*
1426 *Water Management* 280, 108187. doi:10.1016/j.agwat.2023.108187.
- 1427 Reichstein, M., Camps-Valls, G., Stevens, B., Jung, M., Denzler, J., Car-
1428 valhais, N., Prabhat, 2019. Deep learning and process understanding for
1429 data-driven Earth system science. *Nature* 566, 195–204. doi:10.1038/
1430 s41586-019-0912-1.
- 1431 Sekiyama, T., Nagashima, A., 2019. Solar sharing for both food and
1432 clean energy production: Performance of agrivoltaic systems for corn,
1433 a typical shade-intolerant crop. *Environments* 6, 65. doi:10.3390/
1434 environments6060065.
- 1435 Shen, C., Appling, A.P., Gentine, P., Bandai, T., Gupta, H., Tartakovsky, A.,
1436 Baity-Jesi, M., Fenicia, F., Kifer, D., Li, L., Liu, X., Ren, W., Zheng, Y.,
1437 Harman, C.J., Clark, M., Farthing, M., Feng, D., Kumar, P., Aboelyazeed,
1438 D., Rahmani, F., Song, Y., Beck, H.E., Bindas, T., Dwivedi, D., Fang,
1439 K., Höge, M., Rackauckas, C., Mohanty, B., Roy, T., Xu, C., Lawson,
1440 K., 2023. Differentiable modelling to unify machine learning and physical
1441 models for geosciences. *Nature Reviews Earth & Environment* 4, 552–567.
1442 doi:10.1038/s43017-023-00450-9.
- 1443 Siebert, S., Ewert, F., Eyshi Rezaei, E., Kage, H., Graß, R., 2014. Impact
1444 of heat stress on crop yield—on the importance of considering canopy
1445 temperature. *Environmental Research Letters* 9, 044012. doi:10.1088/
1446 1748-9326/9/4/044012.
- 1447 Sparks, A., 2018. nasapower: A nasa power global meteorology, surface solar
1448 energy and climatology data client for r. *Journal of Open Source Software*
1449 3, 1035. doi:10.21105/joss.01035.
- 1450 Trommsdorff, M., Kang, J., Reise, C., Schindele, S., Bopp, G., Ehmann, A.,
1451 Weselek, A., Högy, P., Obergfell, T., 2021. Combining food and energy

1452 production: Design of an agrivoltaic system applied in arable and vegetable
 1453 farming in germany. *Renewable and Sustainable Energy Reviews* 140,
 1454 110694. doi:10.1016/j.rser.2020.110694.

1455 Valle, B., Simonneau, T., Sourd, F., Pechier, P., Hamard, P., Frisson, T.,
 1456 Ryckewaert, M., Christophe, A., 2017. Increasing the total productivity of
 1457 a land by combining mobile photovoltaic panels and food crops. *Applied*
 1458 *Energy* 206, 1495–1507. doi:10.1016/j.apenergy.2017.09.113.

1459 Waha, K., Dietrich, J.P., Portmann, F.T., Siebert, S., Thornton, P.K., Bon-
 1460 deau, A., Herrero, M., 2020. Multiple cropping systems of the world
 1461 and the potential for increasing cropping intensity. *Global Environmental*
 1462 *Change* 64, 102131. doi:10.1016/j.gloenvcha.2020.102131.

1463 Webber, H., White, J.W., Kimball, B.A., Ewert, F., Asseng, S., Eyshi Rezaei,
 1464 E., Pinter, P.J., Hatfield, J.L., Reynolds, M.P., Ababaei, B., Bindi, M.,
 1465 Doltra, J., Ferrise, R., Kage, H., Kassie, B.T., Kersebaum, K.C., Luig,
 1466 A., Olesen, J.E., Semenov, M.A., Stratonovitch, P., Ratjen, A.M., LaM-
 1467 orte, R.L., Leavitt, S.W., Hunsaker, D.J., Wall, G.W., Martre, P., 2018.
 1468 Physical robustness of canopy temperature models for crop heat stress
 1469 simulation across environments and production conditions. *Field Crops*
 1470 *Research* 216, 75–88. doi:10.1016/j.fcr.2017.11.005.

1471 Weselek, A., Bauerle, A., Hartung, J., Zikeli, S., Lewandowski, I., Högy, P.,
 1472 2021. Agrivoltaic system impacts on microclimate and yield of different
 1473 crops within an organic crop rotation in a temperate climate. *Agronomy*
 1474 *for Sustainable Development* 41, 59. doi:10.1007/s13593-021-00714-y.

1475 Weselek, A., Ehmann, A., Zikeli, S., Lewandowski, I., Schindele, S., Högy,
 1476 P., 2019. Agrophotovoltaic systems: applications, challenges, and op-
 1477 portunities. a review. *Agronomy for Sustainable Development* 39, 35.
 1478 doi:10.1007/s13593-019-0581-3.

1479 Willockx, B., Lavaert, C., Cappelle, J., 2023. Performance evaluation of
 1480 vertical bifacial and single-axis tracked agrivoltaic systems on arable land.
 1481 *Renewable Energy* 217, 119181. doi:10.1016/j.renene.2023.119181.

1482 Woli, P., Jones, J.W., Ingram, K.T., Fraisse, C.W., 2012. Agricultural
 1483 reference index for drought (arid). *Agronomy Journal* 104, 287–300.
 1484 doi:10.2134/agronj2011.0286.

- 1485 Yu, Q., You, L., Wood-Sichra, U., Ru, Y., Joglekar, A.K.B., Fritz, S., Xiong,
1486 W., Lu, M., Wu, W., Yang, P., 2020. A cultivated planet in 2010 – part 2:
1487 The global gridded agricultural-production maps. *Earth System Science*
1488 *Data* 12, 3545–3572. doi:10.5194/essd-12-3545-2020.
- 1489 Zhang, L., Geng, X., Ding, H., Cao, K., Wang, L., Chen, H., Deng, L., Wu,
1490 C., Xiao, M., Bao, E., 2026. Multi-objective optimization of photovoltaic
1491 array layouts on farmland via a hybrid algorithm framework for enhanced
1492 agricultural production and energy generation. *Artificial Intelligence in*
1493 *Agriculture* doi:10.1016/j.aiia.2026.09.002.
- 1494 Zhao, C., Liu, B., Piao, S., Wang, X., Lobell, D.B., Huang, Y., Huang, M.,
1495 Yao, Y., Bassu, S., Ciais, P., Durand, J.L., Elliott, J., Ewert, F., Janssens,
1496 I.A., Li, T., Lin, E., Liu, Q., Martre, P., Müller, C., Peng, S., Peñuelas,
1497 J., Ruane, A.C., Wallach, D., Wang, T., Wu, D., Liu, Z., Zhu, Y., Zhu,
1498 Z., Asseng, S., 2017. Temperature increase reduces global yields of major
1499 crops in four independent estimates. *Proceedings of the National Academy*
1500 *of Sciences* 114, 9326–9331. doi:10.1073/pnas.1701762114.
- 1501 Zhao, C., Liu, B., Xiao, L., Hoogenboom, G., Boote, K.J., Kassie, B.T.,
1502 Pavan, W., Shelia, V., Kim, K.S., Hernandez-Ochoa, I.M., Wallach, D.,
1503 Porter, C.H., Stockle, C.O., Zhu, Y., Asseng, S., 2019. A simple crop
1504 model. *European Journal of Agronomy* 104, 97–106. doi:10.1016/j.eja.
1505 2019.01.009.

Table 2: Held-out performance (92 test sites, seasons 2015–2019, canopy cooling $\kappa = 1^\circ\text{C}$). The density (GCR) of each site is the largest value on the grid 0.05, 0.075, ..., 0.5 at which the method’s mean retention in the site’s 2001–2014 seasons is ≥ 0.9 . E : electricity ($\text{MWh ha}^{-1} \text{ yr}^{-1}$); e : electricity relative to a conventional PV plant; r : mean yield retention; r_{prod} : production-weighted retention; C_s and C_{site} : share of seasons with $r \geq 0.9$ and share of sites whose test-season mean $r \geq 0.9$ (%); $\text{LER} = r + e$.

Method	GCR	E	e	r	r_{prod}	C_s	C_{site}	LER
<i>Baseline</i>								
AV static	0.06	217	0.16	0.917	0.914	93	96	1.07
Seasonal sharing	0.17	397	0.28	0.904	0.902	69	72	1.19
Phenology rule	0.17	465	0.33	0.904	0.902	69	75	1.24
Stress rule	0.13	387	0.28	0.907	0.907	76	82	1.18
SALS	0.17	510	0.36	0.905	0.902	70	74	1.27
<i>+1.5 °C</i>								
AV static	0.06	221	0.16	0.916	0.914	93	92	1.08
Seasonal sharing	0.16	393	0.28	0.905	0.903	72	74	1.19
Phenology rule	0.17	478	0.34	0.905	0.902	68	75	1.25
Stress rule	0.12	377	0.27	0.908	0.908	73	77	1.18
SALS	0.17	518	0.37	0.906	0.903	68	75	1.28
<i>+2 °C</i>								
AV static	0.06	223	0.16	0.916	0.914	92	93	1.08
Seasonal sharing	0.17	394	0.28	0.905	0.904	72	76	1.19
Phenology rule	0.17	483	0.35	0.905	0.903	69	74	1.25
Stress rule	0.12	370	0.26	0.909	0.909	75	78	1.17
SALS	0.17	521	0.38	0.907	0.904	69	75	1.28
<i>+3 °C</i>								
AV static	0.06	231	0.17	0.915	0.914	88	89	1.08
Seasonal sharing	0.17	397	0.29	0.906	0.904	69	75	1.19
Phenology rule	0.18	499	0.36	0.905	0.902	67	72	1.26
Stress rule	0.11	366	0.26	0.908	0.908	74	77	1.17
SALS	0.18	540	0.39	0.907	0.904	68	75	1.30

Table 3: LER of the controllers at different yield-retention floors ρ (main split; the historical design step is repeated for each floor).

Method	Baseline climate				+3 °C			
	0.80	0.85	0.90	0.95	0.80	0.85	0.90	0.95
AV static	1.159	1.115	1.072	1.011	1.174	1.128	1.083	1.013
Stress rule	1.380	1.282	1.182	1.077	1.364	1.269	1.172	1.071
Phenology rule	1.467	1.356	1.236	1.108	1.513	1.395	1.265	1.122
SALS (trained at $\rho = 0.9$)	1.505	1.392	1.270	1.141	1.553	1.432	1.297	1.154
SALS conditioned on ρ	1.504	1.391	1.269	1.141	1.553	1.431	1.296	1.154

Table 4: Sensitivity of the controller comparison to the rotation limit of the trackers and to the harvest-index penalty (LER; main split, every sixth held-out pair, 21 pairs, main shade response, history-based design on a GCR grid with step 0.05). SALS was trained for the main setting. Fixed arrays (no tracking; flat and tilted by 30° from the horizontal about the north–south axis) reach LER 1.048 and 1.043 in the same sample, against 1.053 for static trackers.

Setting	Static tracker	Phenology rule	Grid-tuned window	SALS
Main (limit 60°, window 0.45–0.65)	1.053	1.170	1.180	1.189
Rotation limit 45°	1.057	1.138	1.144	1.144
Rotation limit 80°	1.053	1.206	1.219	1.224
Window 0.30–0.50	1.053	1.169	1.176	1.185
Window 0.60–0.80	1.053	1.171	1.181	1.182
No harvest-index penalty	1.109	1.328	1.347	1.353

Table 5: Rules and the learned controller pooled over the three spatial splits (76 distinct crop–site pairs, every third held-out pair; baseline climate; main shade response; history-based design on a GCR grid with step 0.05, coarser than the 0.025 grid of the main analysis). The grid-tuned window shares light while thermal time lies between 0.15 and 1.0 of the season requirement (best of a 4×3 grid on the training-block seasons of each split, the same on all three; the written rule uses 0.05 to 1.0). The CMA-ES schedule has five parameters (start, end, two ramp widths, level) was tuned on each split’s training block and stopped, at the generation cap, at an almost binary window from 0.20 to 0.93 of the requirement at level 1.00 (see Methods).

Method	GCR	E/E_{PV}	Retention	Seasons ≥ 0.9 (%)	Pairs ≥ 0.9 (%)	LER
AV static	0.059	0.153	0.922	95	96	1.076
Seasonal sharing	0.151	0.249	0.915	84	91	1.164
Stress rule	0.118	0.255	0.918	83	88	1.173
Phenology rule (written)	0.154	0.292	0.916	85	91	1.207
Grid-tuned window	0.161	0.317	0.916	84	89	1.233
CMA-ES optimised window	0.158	0.327	0.916	83	87	1.244
SALS	0.161	0.335	0.916	83	87	1.251

Table 6: Simulated relative yield at the shading of published field trials (static arrays, training-block sites). Observed values: rice, single trial; wheat and potato, the worse of the two seasons (the better season gave +2.7% and +11%); maize, mean of the three seasons of the dynamic system (DAV) and single values for the other two systems. The same trials enter the calibration of Section 2.6, so the comparison is not independent.

Trial	Shading	Observed	Conservative	Calibrated
Rice, Gonocruz et al. (2021)	0.25	0.84	0.77	0.79
Wheat, Weselek et al. (2021)	0.30	0.81	0.66	0.74
Potato, Weselek et al. (2021)	0.30	0.82	0.72	0.72
Maize, Ramos-Fuentes et al. (2023) (DAV)	0.35	0.83	0.70	0.70
Maize, Ramos-Fuentes et al. (2023) (half)	0.30	0.91	0.75	0.75
Maize, Ramos-Fuentes et al. (2023) (full)	0.50	0.70	0.54	0.54

Table 7: Sensitivity of the deployable light-sharing results to the crop shade response (rule controllers, baseline climate, floor $\rho = 0.9$, split 0, every fourth held-out site, 22 sites). Conservative: proportional response (main analysis); calibrated: radiation-use compensation fitted to Laub et al. (2022); field-fitted: fitted to the field trials, maize harvest index unaffected.

Shade response	Method	GCR	E/E_{PV}	Retention	LER
conservative	AV static	0.06	0.15	0.92	1.07
	Phenology rule	0.17	0.33	0.90	1.23
	Seasonal sharing	0.16	0.28	0.90	1.18
	Stress rule	0.11	0.24	0.91	1.15
calibrated	AV static	0.07	0.18	0.91	1.09
	Phenology rule	0.19	0.37	0.90	1.28
	Seasonal sharing	0.19	0.32	0.90	1.22
	Stress rule	0.13	0.29	0.91	1.20
field-fitted	AV static	0.16	0.41	0.91	1.32
	Phenology rule	0.37	0.70	0.92	1.62
	Seasonal sharing	0.37	0.61	0.92	1.53
	Stress rule	0.29	0.63	0.90	1.53

Table 8: Predicting relative yield under shade from 428 observations of 58 studies (Laub et al., 2022): leave-one-study-out cross-validation (every study held out once; observations with $s > 0$). Covariates: irradiation, climate zone, shade type, intercropping.

Model	RMSE	MAE	Median AE	Mean MAE per study
Proportional loss, $y = 1 - s$	0.430	0.305	0.210	0.224
One curve for all crops	0.372	0.265	0.193	0.208
Curve per crop type, no pooling	0.313	0.208	0.141	0.165
Curve per crop type, partial pooling	0.312	0.210	0.150	0.162
Partial pooling + covariates	0.311	0.206	0.148	0.161
Gradient-boosted trees	0.321	0.219	0.162	0.184

Table 9: Leave-one-study-out prediction error (RMSE of relative yield, observations with $s > 0$) by crop type: proportional loss, hierarchical curve with partial pooling, and gradient-boosted trees. The last columns give the difference hierarchical minus proportional with a 95% cluster-bootstrap interval over studies (negative: hierarchical better).

Group	Obs./studies	Proportional	Hierarchical	Boosted t
C3 cereals	30/10	0.214	0.187	0.205
Maize	40/10	0.225	0.235	0.248
Grain legumes	58/14	0.215	0.225	0.234
Tubers/root crops	10/2	0.250	0.254	0.257
Four crop types used by the simulator	138/30	0.220	0.222	0.234
Other types (berries, forages, fruits, vegetables)	201/29	0.528	0.361	0.369
All types	339/58	0.430	0.312	0.321

Table 10: Relative yield at 20% and 40% shading: data-driven model (median and 80% bootstrap interval over studies; crop-type curve, covariates at their reference values: mean irradiation, temperate zone, no solar-panel shade, no intercropping) and simulator under the two calibrated ensembles (mean of members; E1 fitted to hand-read curves, E2 fitted to the data-driven curves). The data-driven curve of rice is that of C3 cereals (no rice in the data set). Observations and studies: C3 cereals 41/10, maize 51/10, grain legumes 76/14, tubers 12/2.

Crop	Shading	Data-driven	Simulator, E1	Simulator, E2	Proportional
Wheat	20%	0.93 [0.88, 0.96]	0.88	0.92	0.80
Wheat	40%	0.72 [0.59, 0.78]	0.70	0.77	0.60
Rice	20%	0.93 [0.88, 0.96]	0.85	0.90	0.80
Rice	40%	0.72 [0.59, 0.78]	0.66	0.73	0.60
Maize	20%	0.73 [0.58, 0.86]	0.86	0.90	0.80
Maize	40%	0.42 [0.27, 0.59]	0.69	0.75	0.60
Soybean	20%	0.71 [0.65, 0.79]	0.77	0.74	0.80
Soybean	40%	0.40 [0.34, 0.53]	0.53	0.48	0.60
Potato	20%	0.93 [0.92, 0.97]	0.88	0.92	0.80
Potato	40%	0.71 [0.67, 0.83]	0.73	0.79	0.60

Table 11: Shade-response parameters of ensemble E2, calibrated by gradient descent through the simulator to the data-driven curves and the field trials (12 randomised fits; mean $\pm$ s.d.): radiation-use compensation c and multiplier h on the flowering-window harvest-index penalty (1 = assumed value).

Crop	c	h multiplier
Wheat	0.91 ± 0.31	0.74 ± 0.53
Rice	0.61 ± 0.25	0.83 ± 0.33
Maize	0.14 ± 0.11	0.16 ± 0.13
Soybean	0.05 ± 0.09	1.66 ± 0.38
Potato	0.62 ± 0.20	0.90 ± 0.50

Table 12: Land-equivalent ratio of rules and learned controllers under four true shade responses, with the array density designed under the true response (split 0, baseline climate, floor $\rho = 0.9$, every sixth held-out pair, 21 pairs, coarse GCR grid, one training seed; season means, so differences differ slightly from the pair-level intervals in the text). SALS is trained on the proportional response; robust controllers are trained on random members of ensemble E1, E2 or E3 (adaptive: with the measured biomass ratio). The proportional and harsh responses lie outside the ensembles.

Method	Proportional	Calibrated	Field-fitted	Harsh
AV static	1.052	1.059	1.254	1.037
Seasonal sharing	1.133	1.156	1.385	1.097
Stress rule	1.129	1.160	1.406	1.086
Feedback rule	1.139	1.164	1.392	1.089
Phenology rule (written)	1.164	1.190	1.456	1.122
Grid-tuned window	1.170	1.199	1.472	1.129
CMA-ES optimised window	1.173	1.198	1.479	1.125
SALS	1.178	1.209	1.489	1.132
SALS-robust, E1	1.179	1.204	1.491	1.127
SALS-adaptive, E1	1.176	1.202	1.495	1.124
SALS-robust, E2	1.165	1.192	1.483	1.126
SALS-robust, E3	1.163	1.191	1.481	1.121
SALS-adaptive, E3	1.164	1.191	1.475	1.118

Table 13: Array design under shade-response uncertainty, phenology rule, three spatial splits pooled (76 distinct crop–site pairs, every third held-out pair). Truth: each of five members of the ensemble in turn; the mean response and the member-based rules use the other four members (leave-one-out, so the truth never enters the design). E2: ensemble calibrated to the learned curves and field trials; E3: same procedure with target errors of 0.2 on the learned curves. Pairs: share with mean retention ≥ 0.9 ; shortfall: mean of $\max(0, 0.9 - \text{retention})$ per pair.

Design	E2		LER	E3		
	Pairs ≥ 0.9 (%)	Shortfall		Pairs ≥ 0.9 (%)	Shortfall	
oracle (true response)	87	0.0025	1.486	85	0.0029	1
mean response	70	0.0062	1.493	71	0.0093	1
floor met in 50% of other members	65	0.0084	1.510	67	0.0125	1
floor met in 75% of other members	78	0.0045	1.462	79	0.0083	1
floor met in all other members	89	0.0019	1.412	88	0.0032	1
floor 0.92 under mean response	88	0.0022	1.426	85	0.0051	1
floor 0.94 under mean response	97	0.0007	1.345	92	0.0027	1
floor 0.96 under mean response	99	0.0003	1.251	96	0.0015	1

Table 14: Design rules outside the ensemble: share of crop–site pairs (%) with mean retention ≥ 0.9 under four fixed responses (P proportional, C calibrated to the meta-analysis, F field-fitted, H harsh), phenology rule, three splits pooled (76 distinct pairs). The member-based rules use the five members of E2 or E3, the mean-response design and the margins use the mean of all twelve members, and the oracle uses the true response. LER in the last two columns is the mean over the four responses; the oracle rows are identical for E2 and E3 because the oracle uses the true response.

Design	E2				E3				LER	
	P	C	F	H	P	C	F	H	E2	E3
oracle (true response)	91	89	87	84	91	89	87	84	1.283	1.283
mean response	13	13	68	12	13	17	79	7	1.435	1.448
floor met in 75% of members	16	16	80	12	22	25	93	13	1.384	1.397
floor met in all members	17	17	80	12	29	32	93	16	1.348	1.336
floor 0.92 under mean response	17	20	86	12	29	33	91	20	1.376	1.399
floor 0.94 under mean response	45	47	97	34	45	45	97	29	1.303	1.345
floor 0.96 under mean response	63	78	99	54	58	63	99	51	1.220	1.264

Table 15: Assumed economic parameters (illustrative, not data). Ranges used in the sensitivity analysis are in parentheses.

Parameter	Value
Electricity price	60 USD MWh ⁻¹ (30–120)
Capex, AV single-axis tracker plant	1.0 USD Wp ⁻¹ (0.7–1.5)
Capex, conventional tracker plant	0.75 USD Wp ⁻¹
Discount rate, lifetime, O&M	6% (4–8%), 25 years, 1.5% of capex per year
Installed capacity	2.1 MWp per hectare of module area share g (0.21 kWp m ⁻²)
Crop price per tonne of dry matter	wheat 290, rice 400, maize 250, soybean 520, potato 1000 USD (0.

Table 16: Illustrative annual economics per hectare (76 crop–site pairs, three splits, main shade response, baseline climate; parameters in Table 15). Net value is relative to the open field (crop costs unchanged) in kUSD ha⁻¹ yr⁻¹ at three electricity prices; break-even is the price at which the net value is zero. The conventional plant replaces the crop.

Method	E (MWh)	Capex annuity + O&M	Crop loss	Net at 60	Net at 90
AV static	200	11.5	0.27	0.3	6.3
Seasonal sharing	326	29.6	0.31	-10.3	-0.5
Stress rule	336	23.2	0.30	-3.3	6.8
Phenology rule (written)	382	30.1	0.30	-7.5	4.0
Grid-tuned window	415	31.6	0.31	-7.0	5.5
CMA-ES optimised window	428	31.0	0.30	-5.6	7.3
SALS	438	31.6	0.31	-5.6	7.6
Conventional PV plant	1306	58.7	3.60	16.0	55.2

Table 17: One-at-a-time sensitivity of the net value (kUSD ha⁻¹ yr⁻¹, relative to the open field) at 90 USD MWh⁻¹, densities designed for the floor as in the main analysis. In the capex rows the capex of the conventional plant is set to 0.75 times that of the agrivoltaic plant.

Parameter	Value	Static arrays	Optimised window	SALS	Conventional plant
Base		6.3	7.3	7.6	55.2
AV capex (USD/Wp)	0.7	9.7	16.6	17.1	72.8
AV capex (USD/Wp)	1.5	0.6	-8.2	-8.2	25.8
Discount rate	4%	8.0	12.0	12.4	64.1
Discount rate	8%	4.4	2.1	2.4	45.4
Crop prices	0.5×	6.4	7.4	7.7	57.0
Crop prices	2.0×	6.0	7.0	7.3	51.6
Electricity price	60	0.3	-5.6	-5.6	16.0
Electricity price	120	12.3	20.1	20.7	94.4

Table A.18: Paired comparison of SALS with each baseline on the 92 held-out sites of the main split (unit of analysis: site mean over test seasons). ΔLER : mean difference (SALS minus baseline) with cluster-bootstrap 95% CI; p : two-sided Wilcoxon signed-rank test; % better: share of sites where SALS has the higher LER; Δr : difference in mean yield retention with its Wilcoxon p -value.

Climate	Baseline	ΔLER [95% CI]	p	% better	Δr	p	Sites
Baseline	AV static	+0.205 [+0.176, +0.236]	$< 10^{-4}$	100	-0.011	$< 10^{-4}$	92
Baseline	Seasonal sharing	+0.088 [+0.068, +0.109]	$< 10^{-4}$	97	+0.001	0.922	92
Baseline	Phenology rule	+0.038 [+0.025, +0.053]	$< 10^{-4}$	84	+0.001	0.723	92
Baseline	Stress rule	+0.096 [+0.071, +0.124]	$< 10^{-4}$	93	-0.002	0.055	92
+1.5 °C	AV static	+0.208 [+0.180, +0.237]	$< 10^{-4}$	100	-0.010	$< 10^{-4}$	92
+1.5 °C	Seasonal sharing	+0.095 [+0.077, +0.114]	$< 10^{-4}$	98	+0.001	0.812	92
+1.5 °C	Phenology rule	+0.033 [+0.022, +0.045]	$< 10^{-4}$	86	+0.002	0.855	92
+1.5 °C	Stress rule	+0.108 [+0.083, +0.134]	$< 10^{-4}$	98	-0.001	0.041	92
+2 °C	AV static	+0.209 [+0.181, +0.238]	$< 10^{-4}$	100	-0.010	$< 10^{-4}$	92
+2 °C	Seasonal sharing	+0.097 [+0.079, +0.115]	$< 10^{-4}$	97	+0.001	0.876	92
+2 °C	Phenology rule	+0.032 [+0.022, +0.044]	$< 10^{-4}$	85	+0.002	0.496	92
+2 °C	Stress rule	+0.115 [+0.089, +0.142]	$< 10^{-4}$	98	-0.003	0.016	92
+3 °C	AV static	+0.219 [+0.190, +0.250]	$< 10^{-4}$	100	-0.009	$< 10^{-4}$	92
+3 °C	Seasonal sharing	+0.108 [+0.089, +0.129]	$< 10^{-4}$	96	+0.001	0.935	92
+3 °C	Phenology rule	+0.034 [+0.022, +0.046]	$< 10^{-4}$	86	+0.002	0.330	92
+3 °C	Stress rule	+0.131 [+0.102, +0.162]	$< 10^{-4}$	98	-0.002	0.142	92